MILITARY TECHNICAL ACADEMY

BATTALION 1

FINAL REPORT

SCIENTIFIC RESEARCH PROJECT

Project title: Enhancing the Quality of Thermal Camera Imagery for Improved Visual Observation.

Project code: 25.V4.36

Field of study: Information Technology and Communications

Hanoi, April 2026

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Hanoi, April 2026

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LIST OF ABBREVIATIONS

Abbreviation	Full terminology
CLAHE	Contrast Limited Adaptive Histogram Equalization
MSGF	Multi-Scale Guided Filtering
RGF	Rolling Guidance Filter
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure
RGB	Red, Green, Blue

INTRODUCTION

In recent years, thermal imaging systems have played an increasingly important role in a wide range of applications, including defense and security, surveillance, search and rescue, and monitoring in complex environments. Unlike conventional cameras that capture visible light, thermal cameras operate based on infrared radiation emitted by objects, enabling target detection and observation under low-light conditions or in challenging environments such as nighttime, fog, or heavy rain.

However, images acquired from thermal cameras often suffer from several limitations in visual quality. Due to the characteristics of infrared sensors and acquisition conditions, thermal images typically exhibit a narrow dynamic range, high noise levels, and limited capability in representing structural details. These factors reduce the visual perception of human observers and negatively affect the performance of automated image processing systems.

Various thermal image enhancement methods have been proposed to address these limitations. Traditional approaches such as Rolling Guidance Filter (RGF), Multi-Scale Guided Filter (MSGF), and local contrast enhancement techniques have demonstrated effectiveness in improving image sharpness and highlighting important structures. In addition, some recent approaches exploit information from complementary imaging modalities to further enhance image quality and overall visual perception.

Motivated by practical requirements and the inherent limitations of thermal imagery, this study aims to investigate image processing techniques for improving visual quality and observation capability. The research focuses on traditional image processing methods, which are suitable for efficient implementation in real-world observation systems.

We authors would like to express their sincere gratitude to Colonel Dr. Tong Minh Duc for his dedicated guidance and support throughout the research process. Due to limitations in research experience, the report may still contain shortcomings, and constructive feedback from instructors and reviewers is highly appreciated.

CHAPTER 1: OVERVIEW OF THE RESEARCH PROBLEM

1.1. Research problem

1.1.1. Motivations

Despite the extensive development of thermal image enhancement methods, improving the visual quality of thermal images remains a challenging and unresolved problem. The inherent characteristics of thermal imagery, including low contrast, high noise levels, and limited structural detail representation, make it difficult to effectively highlight objects of interest.

Traditional image processing techniques, such as edge-preserving filtering, multi-scale decomposition, and contrast enhancement, have been widely used to improve specific aspects of thermal images. However, when applied individually, these methods often fail to balance detail enhancement and noise suppression. Some approaches emphasize dominant structures at the cost of fine details, while others amplify noise or introduce instability in the image background.

Deep learning-based methods have achieved strong performance in various image processing tasks. However, they typically require large-scale training data and high computational resources, which limit their deployment in real-world observation systems with constrained resources.

Therefore, developing thermal image enhancement methods based on traditional techniques, while achieving better control over detail enhancement and noise suppression, remains an important research direction.

1.1.2. Research objectives

This study aims to develop a thermal image enhancement method that improves visual quality and enhances the visibility of important structural details in thermal imagery. In addition to improving image contrast and detail representation, the study also aims to address the trade-off between detail enhancement and noise suppression, which remains a key challenge in thermal image processing.

To achieve these objectives, the study focuses on the following specific tasks:

- To review and analyze existing thermal image enhancement methods, as well as related image processing techniques such as guided filtering, multi-scale decomposition, and local detail enhancement, in order to identify their strengths and limitations.
- To develop a structured image processing pipeline that integrates multiple techniques to enhance thermal image quality, including contrast enhancement, structural detail preservation, and noise control.
- To design and implement an experimental framework for evaluating the proposed method using both quantitative metrics and qualitative visual analysis.
- To analyze the performance of the proposed method in comparison with selected existing approaches, thereby assessing its effectiveness and practical applicability in real-world observation scenarios.

1.1.3. Research methodology

To achieve the defined objectives, this study adopts a combination of complementary research methods, ensuring both theoretical rigor and practical validation:

- Theoretical analysis: A comprehensive review and analysis of scientific literature on thermal image enhancement and image processing techniques in computer vision is conducted. This step focuses on identifying key challenges, commonly used approaches, and existing limitations. The insights obtained provide a foundation for selecting appropriate techniques and defining the overall research direction.
- Algorithm design and modeling: Based on the theoretical findings, an image processing model is developed using a multi-scale decomposition framework combined with detail enhancement strategies. The model is designed with consideration of the specific characteristics of thermal imagery, aiming to improve structural representation while maintaining stability and reducing noise.
- Experimental method: The proposed algorithm is implemented and evaluated on thermal image datasets. The experimental process allows direct

observation of enhancement results and supports iterative parameter tuning. This step plays a crucial role in validating the effectiveness of the method under practical conditions.

- Comparative evaluation: Quantitative image quality metrics are employed to assess key characteristics of enhanced images, including information content, structural sharpness, and contrast discrimination. The performance of the proposed method is compared with selected existing methods to provide a comprehensive evaluation of its effectiveness, robustness, and suitability for real-world deployment.

1.2. Research scope

The scope of this study encompasses both the input image data and the image processing methods employed to extract and enhance information from these images. Specifically, the research considers thermal images, visible (RGB) images, and image processing algorithms designed to improve visual quality and perceptual clarity.

1.2.1. Overview of Thermal images

A thermal image is acquired using infrared sensors that capture the thermal radiation emitted by objects within a scene. Unlike conventional imaging systems operating in the visible spectrum, thermal cameras function in the infrared spectrum and represent the temperature distribution of objects in the observed environment.

From a data representation perspective, a thermal image is typically modeled as a single-channel grayscale function:

$$T(x, y)$$

where:

- $T(x, y)$: Thermal intensity at pixel (x, y)
- x, y: Spatial coordinates of the pixel
- $T(x, y) \in [0, L - 1]$: Intensity range with $L = 256$

Due to the characteristics of infrared sensing and acquisition conditions, thermal images exhibit several distinctive properties compared to visible images. In particular, they are characterized by low contrast, relatively high noise levels, and limited structural detail representation. Consequently, objects may be difficult to distinguish from the background, and important geometric features are often not clearly defined.

In this study, thermal images constitute the primary research subject and serve as the main input to the enhancement process. The objective is to improve their visual quality and observation capability through appropriate image processing techniques.

Figure 1.1: Example of a thermal image

These characteristics highlight the inherent limitations of thermal imagery in terms of contrast and structural representation. Accordingly, effective enhancement techniques are required to improve target visibility, forming the foundation for the methods presented in subsequent chapters.

1.2.2. Overview of Visual images

A visible image, also referred to as an RGB image, is acquired using cameras operating in the visible light spectrum. Unlike thermal images, which contain a single intensity channel, RGB images are represented using three channels corresponding to the primary color components: Red (R), Green (G),

and Blue (B). The combination of these three channels enables rich color representation of the observed scene.

Each pixel in an RGB image is represented as a vector of three intensity values:

$$V(x, y) = [\, R(x, y), G(x, y), B(x, y) \,]$$

where:

- $R(x, y)$: Intensity of the Red channel
- $G(x, y)$: Intensity of the Green channel
- $B(x, y)$: Intensity of the Blue channel
- $R, G, B \in \mathbb{R}^{m \times n}$ with intensity values in the range $\in [0, L - 1]$

Thanks to the combination of three color channels, RGB images provide detailed information about object shape, texture, and color. In multi-sensor observation systems, thermal and RGB images are often captured simultaneously using two aligned cameras to ensure spatial and temporal correspondence between the two data sources. This enables the exploitation of complementary information from both modalities to improve overall observation performance.

Within the scope of this study, RGB images are not the primary processing target but serve as a complementary information source to support the enhancement of thermal imagery. Their rich visual representation is leveraged during the image fusion stage to improve overall perceptual quality and target visibility.

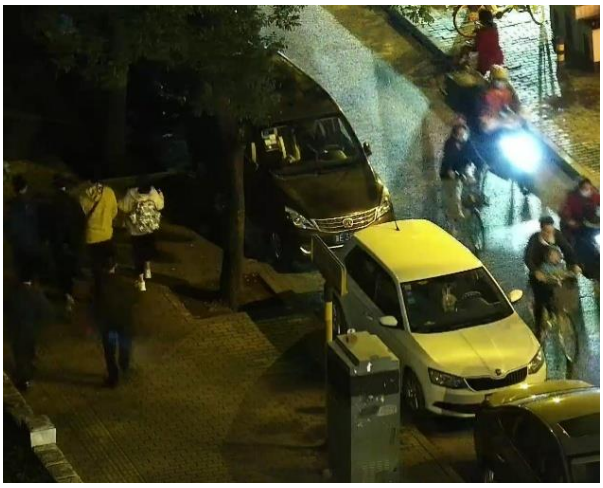


Figure 1.2: Example of a visible (RGB) image

1.2.3. Image processing algorithms

1.2.3.1. Overview of Image Processing

In digital image processing, algorithms are employed to improve visual quality or to extract meaningful information from image data. In the context of image enhancement, the primary objective is to define an appropriate transformation that improves key image characteristics such as contrast, sharpness, and structural detail representation.

For thermal imaging systems, image processing algorithms play a crucial role in addressing inherent limitations of thermal images, including high noise levels, low contrast, and insufficient structural detail. The image enhancement process can be modeled as a transformation from an input image to an output image through a processing function:

$$I_{out}(x, y) = F(I_{in}(x, y))$$

where:

- $I_{in}(x, y)$: Input image
- $I_{out}(x, y)$: Processed output image
- $F(\ldots)$: Image processing function or algorithm

1.2.3.2. Techniques and Algorithms

Within the scope of this study, traditional image processing algorithms are selected as the primary tools for developing the thermal image enhancement method. Techniques such as guided filtering, multi-scale decomposition, and contrast enhancement are combined to form a structured image processing pipeline, where each component addresses specific characteristics of thermal imagery. Guided filtering performs edge-preserving smoothing, multi-scale decomposition separates structural information at different scales, and contrast enhancement improves the visibility of important details. These techniques are directly integrated into the proposed method to achieve controlled enhancement

of structural details while suppressing noise, thereby improving overall visual quality and observation performance.

a. Image Filtering techniques

Image filtering is a fundamental group of methods in digital image processing, used for noise reduction, smoothing, and enhancement of important image details. In thermal image processing, filtering techniques must effectively suppress noise while preserving structural edges of objects.

Traditional linear filters, such as mean and Gaussian filters, can reduce noise effectively but often blur important edge details. Therefore, more advanced edge-preserving methods have been developed, among which guided filtering is a prominent approach. A representative method in this category is the Rolling Guidance Filter (RGF), which removes small-scale noise while maintaining important structural edges.

b. Multi-Scale Image Decomposition

Multi-scale decomposition is a widely used approach in image enhancement. The core idea is to decompose an image into multiple components at different spatial scales, including a base layer and several detail layers.

This process can be expressed as:

$$I = B + \sum_{i=1}^{n} D_i$$

where:

- I: Original image
- B: Base layer representing the overall intensity structure
- D_i: Detail layers at different spatial scales

In thermal image enhancement, multi-scale decomposition enables the separation of structural details from the background, allowing more precise control over the enhancement process. By adjusting each detail layer

independently, important features can be enhanced while maintaining the stability of the base layer.

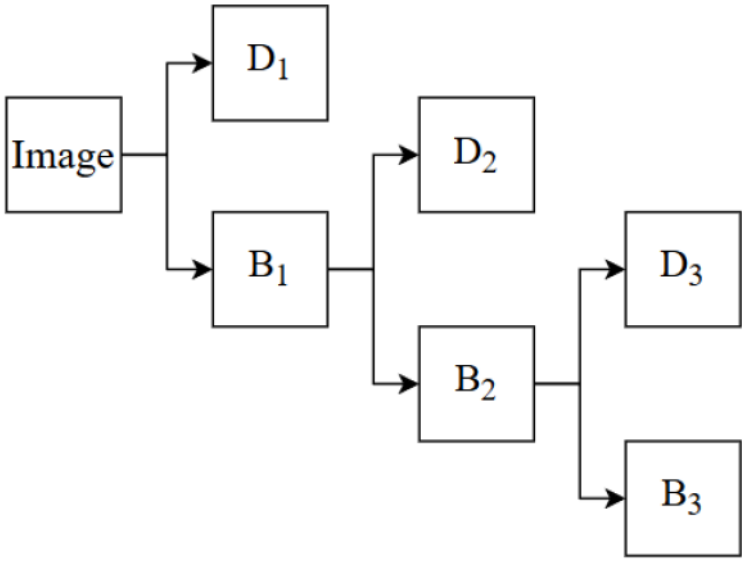


Figure 1.3: Working principle of the multi-scale decomposition method

c. Contrast Enhancement

Improving contrast is a key objective in thermal image enhancement, as it directly affects the visibility of objects within the image. Various contrast enhancement methods have been developed in digital image processing, among which histogram-based techniques are widely used.

A commonly applied method is Contrast Limited Adaptive Histogram Equalization (CLAHE). This approach divides the image into small regions and performs local histogram equalization while limiting the amplification factor to prevent excessive noise enhancement.

Compared to global histogram equalization methods, CLAHE improves local contrast while maintaining overall visual stability. Furthermore, traditional image processing techniques ensure reasonable computational cost, making them suitable for deployment in real-world observation systems.

The combination of image filtering, multi-scale decomposition, and contrast enhancement provides the foundation for constructing an effective image

processing pipeline, aimed at improving visual quality and observation capability of thermal imagery.

1.3. Research scope and Limitations

Defining the research scope is essential for clarifying the objectives and boundaries of the study. In this work, the focus is placed on the preprocessing stage, aiming to enhance the quality of input thermal images for observation systems. The techniques presented in this chapter serve not only as the theoretical foundation but are also directly incorporated into the proposed method in Chapter 2. The overall objective is to develop an image enhancement pipeline that effectively balances detail enhancement and noise suppression, while maintaining computational efficiency for practical deployment.

1.3.1. Research content

This study focuses on investigating image processing methods for enhancing the quality of thermal images acquired from thermal cameras. The main research activities include:

- Reviewing existing thermal image enhancement methods reported in the literature.
- Studying suitable image processing techniques for improving thermal image quality.
- Developing an image processing pipeline to enhance image contrast and improve visual distinguishability.

1.3.2. Data and System scope

The dataset used in this study consists primarily of paired thermal and visible (RGB) images acquired from observation systems, ensuring spatial and temporal correspondence between the two modalities. Thermal images serve as the primary input for the enhancement process, while RGB images provide complementary information to support improved scene interpretation. The effectiveness of the proposed method is evaluated using quantitative image quality metrics, enabling objective assessment of improvements in visual quality after enhancement.

1.3.3. Research limitations

This study is limited to the problem of thermal image enhancement with the objective of improving visual observation quality. Other computer vision tasks, such as object detection, tracking, and recognition, are beyond the scope of this work.

Furthermore, the study focuses exclusively on traditional image processing techniques and does not consider deep learning-based approaches or advanced machine learning models. This constraint is intentional, as the objective is to develop a method with moderate computational cost, suitable for practical deployment in real-world observation systems.

1.4. Related Works

Thermal image enhancement and multi-sensor image fusion have attracted significant attention in the fields of image processing and computer vision. Thermal imagery plays a critical role in various observation systems, particularly in applications such as nighttime surveillance, search and rescue, and defense. However, due to the characteristics of infrared sensors, thermal images often suffer from low contrast, high noise levels, and limited structural detail representation.

To address these limitations, numerous methods have been proposed to enhance thermal image quality or improve overall observation capability. Existing research can be broadly categorized into three main groups: (i) review studies and multi-sensor fusion systems, (ii) traditional image processing methods, and (iii) deep learning-based approaches.

1.4.1. Review studies and Multi-Sensor Observation Systems

One major research direction in thermal imaging focuses on exploiting complementary information from multiple sensors to improve observation performance. In modern observation systems, thermal images are often combined with visible (RGB) images to leverage the strengths of both modalities. While thermal images capture temperature distribution, visible images provide detailed structural and color information of the scene.

Kaur et al. [5] presented a comprehensive survey of image fusion techniques in digital image processing. Their study categorizes fusion methods into several groups, including multi-scale transform-based methods, feature-based methods, and machine learning-based approaches. The survey highlights that combining information from multiple image sources significantly improves the representation quality of the fused output.

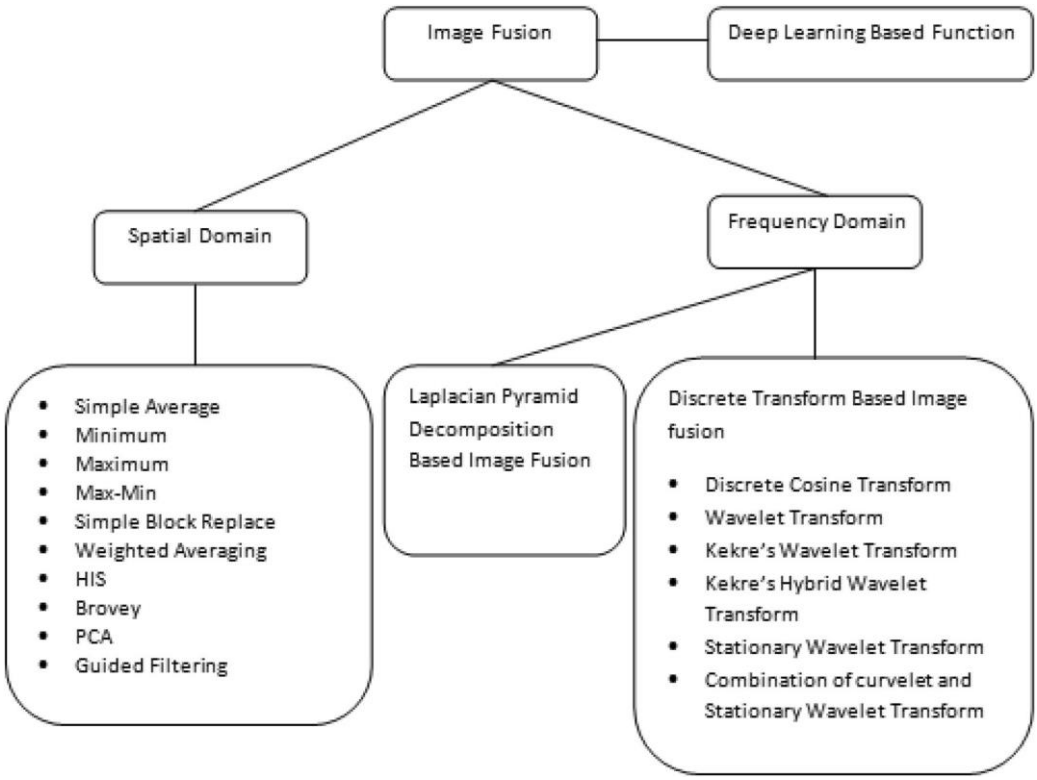


Figure 1.4: Overview of image fusion methods [5]

Li et al. [7] investigated multi-sensor image fusion systems and analyzed methods used in modern observation platforms. Their results indicate that combining thermal and visible data enhances observation performance under low-light or complex environmental conditions.

Ma et al. [16] conducted a detailed survey of thermal-visible image fusion techniques. Their work demonstrates that integrating thermal and color information significantly improves the structural representation of the observed scene.

In addition, Toet and Hogervorst [8] studied night vision systems that incorporate color information to improve object perception in low-light

conditions. Their findings show that augmenting thermal imagery with color cues enhances human visual perception.

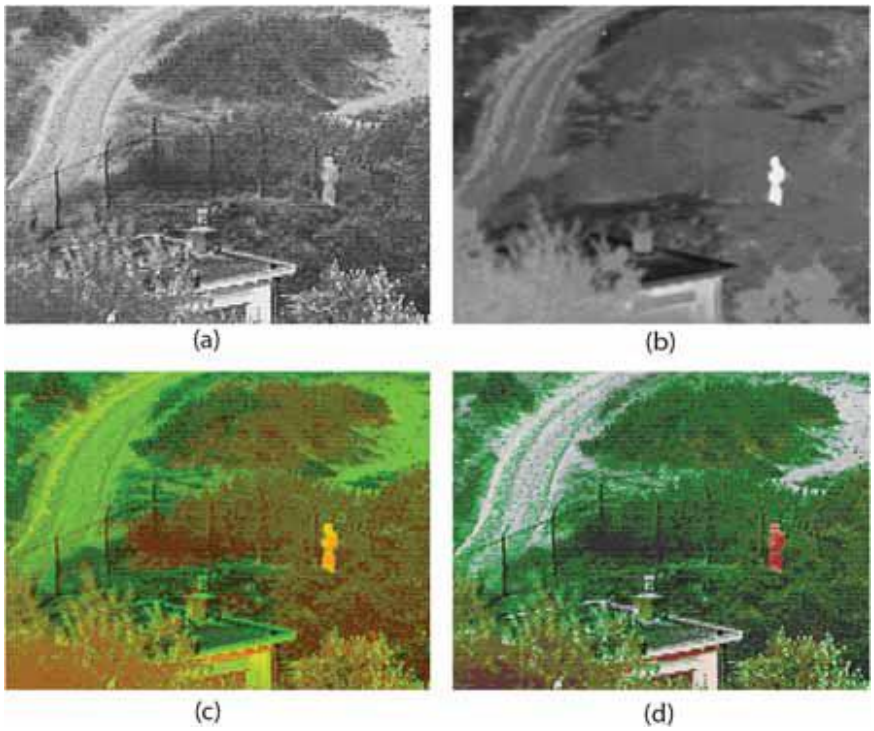


Figure 1.5: Results demonstrating improved visual observation [8]

Although these studies provide valuable insights into fusion techniques and multi-sensor observation systems, most focus on system-level analysis or general fusion frameworks. Limited attention has been given to developing computationally efficient thermal image enhancement algorithms tailored for practical deployment, which motivates the approach proposed in this study.

1.4.2. Traditional Image Processing Methods

In addition to review studies and multi-sensor observation systems, many works have focused directly on developing image processing methods to improve the visual quality of thermal images. This group of approaches is particularly relevant to this study, as the primary objective is to enhance thermal imagery acquired from thermal cameras. In general, traditional approaches are based on three main categories: edge-preserving filtering, multi-scale decomposition, and contrast enhancement.

One of the fundamental methods in this domain is the Guided Image Filter proposed by He et al. [6]. This method performs image smoothing while preserving important structural edges. Compared to classical linear filters such as Gaussian or mean filtering, guided filtering is more suitable for thermal image enhancement as it reduces noise without blurring edge structures. Due to this property, it has become a foundation for many subsequent studies in both image enhancement and image fusion.

Building upon this principle, Li et al. [1] proposed a thermal image enhancement method based on multi-scale guided filtering. Instead of processing the image as a single entity, the image is decomposed into multiple layers at different spatial scales. This multi-scale approach enables selective enhancement of structural details, improving contrast and structural representation. Their results demonstrate that multi-scale features are particularly effective for thermal imagery, especially when object details and background information are distributed across different spatial scales. However, insufficient control between layers may lead to the suppression of fine details or blending with the background after enhancement.

In a different approach, Liu et al. [2] combined Rolling Guidance Filter (RGF) with CLAHE, targeting efficient implementation on FPGA platforms. Compared to standard guided filtering, RGF iteratively removes small-scale structures, thereby emphasizing dominant image features. The integration of CLAHE further improves local contrast and visual clarity. A key strength of this work lies in its focus on hardware implementation and processing efficiency. However, the iterative smoothing characteristic of RGF may weaken fine details or subtle background structures, particularly in low-contrast thermal images.

Beyond direct enhancement methods, several traditional approaches address the problem through image fusion. Toet [10] proposed a low-pass pyramid-based fusion method, where image information is processed at multiple resolution levels before fusion. This approach laid the foundation for subsequent

multi-scale fusion techniques by demonstrating that separating information across frequency bands helps preserve important features from different sources.

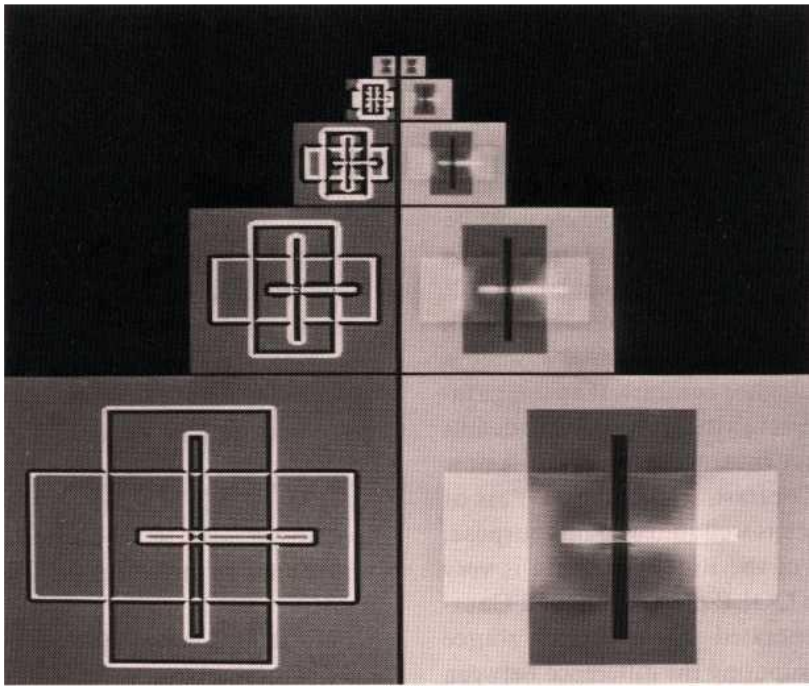


Figure 1.6: Results of the low-pass pyramid method [10]

Zhang et al. [13] introduced a thermal-visible fusion method based on mutual guided filtering, which improves structural consistency in the fused output. However, its effectiveness depends on the structural compatibility between input modalities. Similarly, Zhang et al. [18] proposed an FPGA-based fusion system combining guided filtering with saliency detection, demonstrating the potential of traditional methods for real-time applications. Nevertheless, such fusion-based approaches primarily aim to integrate multiple data sources rather than directly enhance the quality of thermal images.

In addition, Qian et al. [11] proposed an entropy-based enhancement method, where local entropy maps guide the enhancement process. This approach is effective in highlighting low-contrast regions and weak structural details. The work of Le Khanh Thanh et al. [12] also indicates that incorporating complementary information can improve output quality. Although not fully aligned with the scope of this study, these works suggest that content-adaptive enhancement strategies are a promising direction.

Overall, traditional image processing methods offer advantages in terms of simplicity, interpretability, low computational cost, and suitability for real-world deployment. These characteristics make them particularly relevant for thermal image enhancement in resource-constrained systems. However, a fundamental challenge remains unresolved: achieving an effective balance between detail enhancement and noise suppression. Some methods preserve dominant structures at the expense of fine details, while others enhance contrast but introduce noise or background instability. This limitation highlights a critical research gap and motivates the development of improved enhancement methods for thermal imagery.

1.4.3. Deep Learning-based Methods

In parallel with traditional image processing approaches, deep learning-based methods have seen rapid development in recent years and have been widely applied to image enhancement and thermal image fusion tasks. Unlike traditional methods that explicitly define processing steps such as filtering, decomposition, or contrast enhancement, deep learning models learn feature representations directly from data, enabling the modeling of complex nonlinear relationships that are difficult to capture using handcrafted techniques.

DenseFuse, proposed by Li and Wu [4], is a representative method in this category. It employs a dense connection architecture to extract features from thermal and visible images, followed by fusion in the feature domain. DenseFuse demonstrates strong performance in preserving useful information from both modalities and producing visually enhanced outputs. Compared to traditional fusion methods, it benefits from automatic feature learning rather than relying entirely on manually designed rules. However, its performance depends heavily on training data quality, network design, and computational resources.

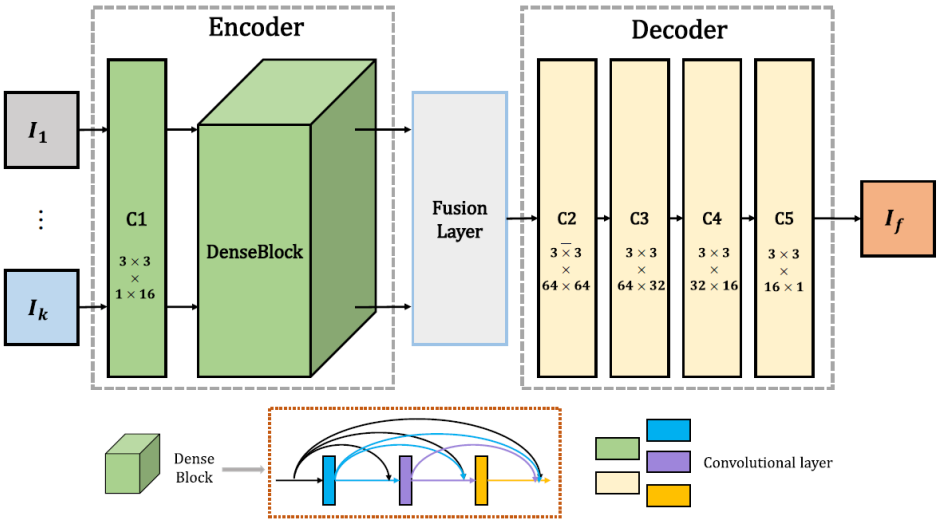


Figure 1.7: Working principle of the DenseFuse method [4]

Another approach, U2Fusion proposed by Xu et al. [9], introduces an unsupervised learning framework for image fusion, eliminating the need for labeled training data. This is particularly advantageous in fusion tasks where ground-truth data is difficult to obtain. The method leverages intrinsic data structures to guide the fusion process. Nevertheless, it still requires substantial training effort and remains computationally demanding.

In terms of output quality, deep learning methods often produce visually superior results due to their ability to learn rich feature representations and model complex relationships between modalities. However, most of these approaches are designed for multi-modal image fusion, whereas the primary objective of this study is to enhance the quality of thermal images themselves. Therefore, while deep learning methods provide valuable insights and benchmarks, they are not the central focus of this work.

Moreover, deep learning approaches present significant challenges for deployment in resource-constrained systems. They typically require large-scale training datasets, extensive training time, high-performance hardware such as GPUs, and complex optimization for real-world implementation. These limitations reduce their suitability for applications that prioritize stability, controllability, and efficient deployment.

As a result, deep learning-based methods are considered primarily as reference approaches for comparison and for defining the boundaries of the selected research direction, rather than as the main implementation strategy in this study.

1.4.4. Method comparisons

Based on the reviewed studies, various approaches have been proposed to improve thermal image quality and enhance observation performance in computer vision systems. However, each category of methods presents inherent limitations, particularly in balancing detail enhancement, noise suppression, and computational cost.

Traditional methods are generally efficient and suitable for real-time or resource-constrained systems, but often struggle to preserve fine details while suppressing noise. Deep learning-based approaches typically achieve superior visual quality, but require significant training data, computational resources, and complex deployment processes. Fusion-based methods improve overall scene representation by combining multiple modalities, but do not directly address the enhancement of thermal images themselves.

Method	Category	Advantages	Limitations	Suitability
MSGF [1]	Traditional	Preserves edges	Loss of fine details	High
RGF + CLAHE [2]	Traditional	Real-time capability	Sensitive to parameter settings	High
DenseFuse [4]	Deep learning	High visual quality	High computational costs	Low
U2Fusion [9]	Deep learning	Unsupervised fusion	High training costs	Low
Mutual guided filtering [13]	Traditional	Stable fusion performance	Depends on structural alignment	Medium

Table 1: Advantages and limitations of selected methods

1.5. Image quality metrics

Evaluating the effectiveness of image processing methods is essential for determining the level of improvement achieved after enhancement. In addition to qualitative visual assessment, various quantitative metrics have been proposed to provide objective evaluation of image quality.

In general image processing tasks, commonly used metrics include Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). These metrics are widely applied in applications such as image restoration, compression, and denoising, where a high-quality reference image is available. In such cases, PSNR and SSIM effectively measure the similarity between the processed image and the reference.

However, in thermal image enhancement, reference-based metrics face significant limitations due to the absence of ground-truth thermal images. Thermal data is influenced by multiple factors, including environmental conditions, sensor characteristics, and observation distance. As a result, metrics such as PSNR do not accurately reflect perceptual improvement, while SSIM primarily measures structural similarity and does not fully capture enhancements in visual clarity or target visibility.

Given these constraints and the practical requirements of observation systems, this study adopts no-reference image quality metrics that directly reflect perceptual characteristics of enhanced images. The selected metrics include:

- Image Entropy
- Sobel Energy
- Laplacian Variance
- RMS Contrast

These metrics allow evaluation of image quality from multiple perspectives, including information content, structural sharpness, and contrast discrimination.

During evaluation, all metrics are computed on 8-bit grayscale images, with input images converted to grayscale and normalized to the range [0, 255] when necessary.

1.5.1. Image Entropy

Image entropy is a widely used metric in image processing for evaluating the richness of information within an image. It reflects the distribution and diversity of grayscale intensity levels, thereby indicating the amount of visual information present. Entropy is defined based on Shannon's formulation:

$$\boldsymbol{H(I_8) = -\sum_{0}^{255} p(i) \log_2 p(i)} \ \ [12]$$

where:

- $p(i)$: Probability of occurrence of grayscale level i
- i: Grayscale intensity level ∈ [0, 255]

A higher entropy value indicates a more diverse intensity distribution and, consequently, greater information content. In thermal images, entropy is particularly useful for assessing improvements in low-contrast regions and background details. However, a higher entropy does not necessarily imply better visual quality, as noise can also increase entropy. Therefore, this metric should be interpreted in conjunction with other measures.

1.5.2. Sobel energy

The Sobel operator is a standard method for edge detection in images. In thermal image analysis, Sobel energy is used as a quantitative measure of structural sharpness. It is computed as the average gradient energy over the image:

$$\boldsymbol{S(I_8) = \frac{1}{N} \sum_{x,y} \left(G_x(x,y)^2 + G_y(x,y)^2 \right)}$$

where:

- G_x, G_y: Horizontal and vertical Sobel responses
- N: Total number of pixels

A higher Sobel energy value indicates stronger edge responses and more pronounced structural details. This metric is particularly effective for evaluating the enhancement of object boundaries and edge clarity.

1.5.3. Laplacian Variance

Laplacian variance is a commonly used metric for assessing overall image sharpness. It is based on the Laplacian operator, which is sensitive to high-frequency components such as edges. The variance is defined as:

$$\boldsymbol{Var_{Laplacian}(I_8) = \frac{1}{N}\sum_{x,y}(L(x,y) - \mu_L)^2}$$

where:

- $L(x, y)$: Laplacian response at pixel (x, y)
- μ_L: Mean value of the Laplacian image
- N: Total number of pixels

Higher variance values indicate a greater presence of high-frequency components, corresponding to sharper edges and finer structural details. This metric is effective for evaluating improvements in image sharpness after enhancement.

1.5.4. RMS Contrast

Root Mean Square (RMS) contrast measures the dispersion of intensity values in an image, reflecting the overall contrast between dark and bright regions. It is defined as:

$$C_{RMS}(I_8) = \sqrt{\frac{1}{N}\sum_{x,y}(I_8(x,y) - \mu)^2}$$

where:

- $I_8(x, y)$: Intensity value at pixel (x, y)
- μ: Mean intensity of the image
- N: Total number of pixels

In thermal images, the temperature difference between objects and the background is often limited. Therefore, improving contrast plays a critical role in enhancing visual distinguishability. RMS contrast provides a quantitative measure of this improvement.

1.5.5. Overall evaluation

The combination of the above metrics enables a comprehensive evaluation of enhanced image quality across multiple aspects, including information content, structural sharpness, and contrast discrimination. This multi-metric approach provides a more reliable and balanced assessment compared to relying on a single indicator.

From the preceding analysis, it is evident that although numerous methods have been proposed for thermal image enhancement, the challenge of balancing detail enhancement and noise suppression remains unresolved under practical observation conditions. Motivated by this research gap, this study proposes a thermal image enhancement pipeline based on multi-scale decomposition combined with selective detail enhancement, aiming to improve structural representation and overall visual quality.

The main contributions of this study are summarized as follows:

- A multi-scale decomposition framework based on the Rolling Guidance Filter (RGF) to effectively separate base and detail layers across different spatial scales.
- An adaptive detail enhancement mechanism guided by local image characteristics to control the enhancement process.

- A reconstruction strategy that balances contrast, sharpness, and visual stability

Based on this foundation, Chapter 2 presents the proposed methods and the detailed image processing pipeline.

CHAPTER 2: RELATED WORKS AND PROPOSED METHOD

2.1. Fundamental Image Processing methods

Based on the review of representative approaches in thermal image processing and multi-sensor image fusion, it can be observed that traditional image processing methods—particularly edge-preserving filtering and multi-scale decomposition—continue to play a significant role in improving thermal image observation. This is primarily due to their reasonable computational cost and suitability for deployment in practical systems.

However, existing methods still exhibit notable limitations, especially in balancing noise suppression, edge preservation, and the retention of low-contrast details. Some approaches achieve effective smoothing but tend to suppress fine details, while others enhance structural features at the cost of amplifying noise or introducing instability in the background.

Motivated by these observations, the principles of key techniques such as the Rolling Guidance Filter and multi-scale decomposition methods are analyzed to establish the theoretical foundation for the image enhancement process. Based on this foundation, this study proposes a thermal image enhancement pipeline that integrates multi-scale decomposition with selective detail enhancement, enabling improved control over the interaction between the base layer and detail layers.

Furthermore, to support visual observation in multi-sensor systems, this chapter also presents an approach for integrating enhanced thermal image information with RGB images through the luminance component. This strategy leverages the ability of thermal imagery to highlight targets while preserving the natural color representation of the observed scene.

2.1.1. Rolling Guidance Filter

The Rolling Guidance Filter (RGF) is an edge-preserving filtering algorithm designed to perform controlled image smoothing through an iterative mechanism. The key characteristic of RGF is that the output of each iteration is

used as the guidance image for the subsequent filtering step. Through this iterative update process, small-scale details and noise are progressively removed, while important structural edges are preserved.

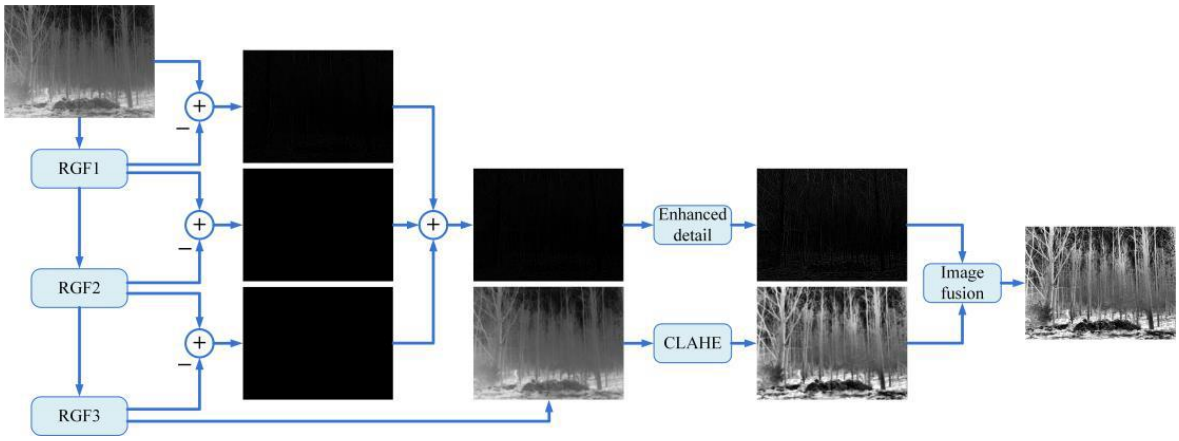


Figure 2.1: Working principle of the RGF method [2]

For thermal image processing, RGF offers several advantages. Due to the characteristics of infrared sensors and acquisition conditions, thermal images often contain background noise and regions with relatively uniform intensity. The application of RGF helps smooth these regions while maintaining the main structural boundaries of objects, thereby improving visual stability.

However, as an iterative smoothing process, RGF also exhibits certain limitations. When the number of iterations or the smoothing strength increases, fine details may be weakened or completely removed. This effect is particularly noticeable for weak background structures or objects with low contrast relative to their surroundings. As a result, the processed image becomes smoother but may lose important structural information necessary for visual interpretation.

2.1.2. Multi-Scale Guided Filtering

Multi-Scale Guided Filtering (MSGF) is an image enhancement method based on the principle of multi-scale image decomposition. Built upon the guided filtering framework, this approach decomposes an image into multiple components at different spatial scales.

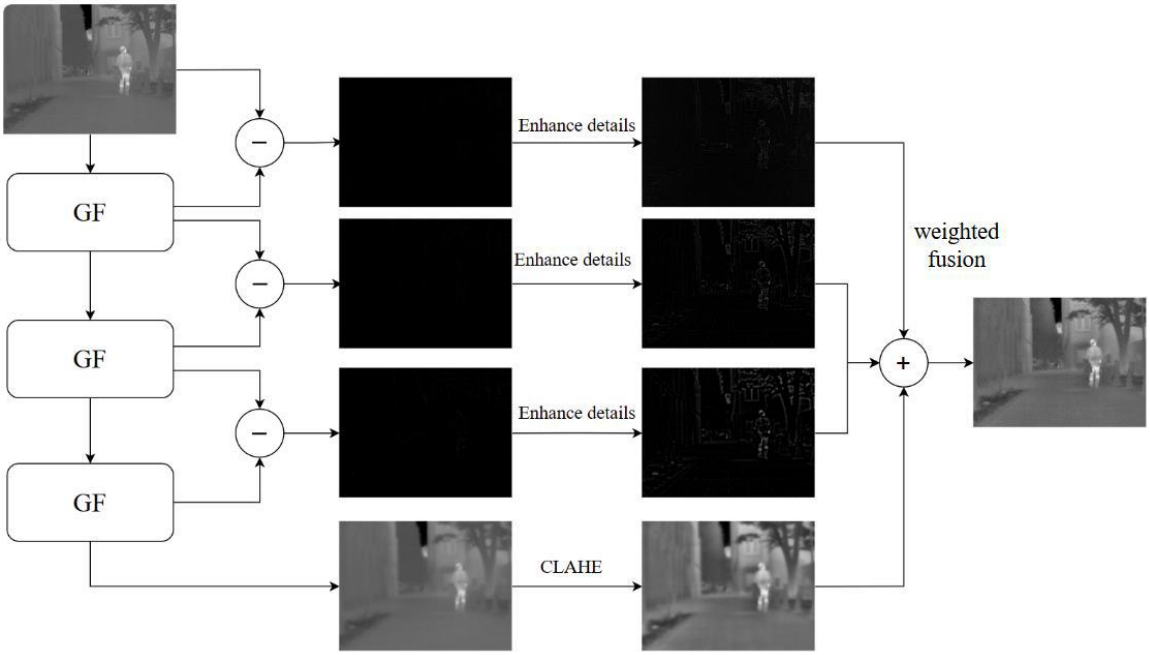


Figure 2.2: Working principle of the MSGF method [1]

Typically, the input image is decomposed into a base layer, representing the overall intensity distribution, and several detail layers corresponding to different spatial scales. In many implementations, these detail layers are categorized into three levels: fine details, intermediate details, and large-scale structural components. This decomposition enables the application of different enhancement strategies to each type of detail. For thermal images, the multi-scale approach allows effective enhancement of salient structures such as humans or vehicles within the observed scene. Experimental results of MSGF demonstrate significant improvements in contrast and structural representation across various scenarios.

Nevertheless, MSGF also presents certain limitations. When enhancement is strongly focused on dominant detail layers, less prominent background details may be suppressed or lost. This issue becomes more pronounced in complex scenes containing multiple objects or varying structural patterns. In addition, the parameters of multi-scale processing are typically fixed across the image, which may not adequately reflect the structural variability of thermal images under different observation conditions.

2.1.3. Overall evaluation

From the above analysis, both RGF and MSGF provide significant advantages for thermal image enhancement. RGF effectively reduces noise and performs smoothing while preserving major structural edges, whereas MSGF enables the extraction and enhancement of image details across multiple spatial scales, thereby highlighting important objects in the scene.

However, when applied independently, each method exhibits inherent limitations related to the preservation of background details and the balance of overall image information. RGF may suppress fine structures, while MSGF may overemphasize dominant features at the expense of weaker details. Therefore, developing a method that combines the strengths of these approaches while mitigating their respective limitations represents a necessary and promising research direction.

2.2. Proposed method

This study proposes a thermal image enhancement pipeline based on traditional image processing techniques. The objective is to improve the observability of thermal images by enhancing structural details, increasing contrast, and suppressing noise, while maintaining computational efficiency for real-world deployment.

2.2.1. Overall pipeline

The proposed method consists of two stages. The first stage performs thermal image enhancement and represents the core of this study. The second stage utilizes the enhanced thermal image to support observation in multi-sensor systems through luminance-based integration with RGB images.

a. Phase 1:

The input thermal image is processed using a multi-scale decomposition framework to separate the base layer and detail layers at different spatial scales. This separation enables independent control of background smoothing and detail

enhancement. Direct global enhancement often leads to noise amplification or loss of low-contrast details. Therefore, decomposition is used to isolate different information components and allow targeted processing

The detail layers are selectively enhanced based on local image characteristics, specifically local entropy and local contrast. Regions with meaningful structural information are strengthened, while regions prone to noise are controlled. The base layer is adjusted using controlled contrast enhancement to improve overall intensity distribution while maintaining stability. The output of this stage is an enhanced thermal image with improved structure, contrast, and visual consistency.

b. Phase 2:

The second stage integrates the enhanced thermal image with RGB data to improve observation in multi-sensor systems. The RGB image is treated as a complementary source of visual information. The second stage integrates the enhanced thermal image with RGB data to improve observation in multi-sensor systems. The RGB image is treated as a complementary source of visual information

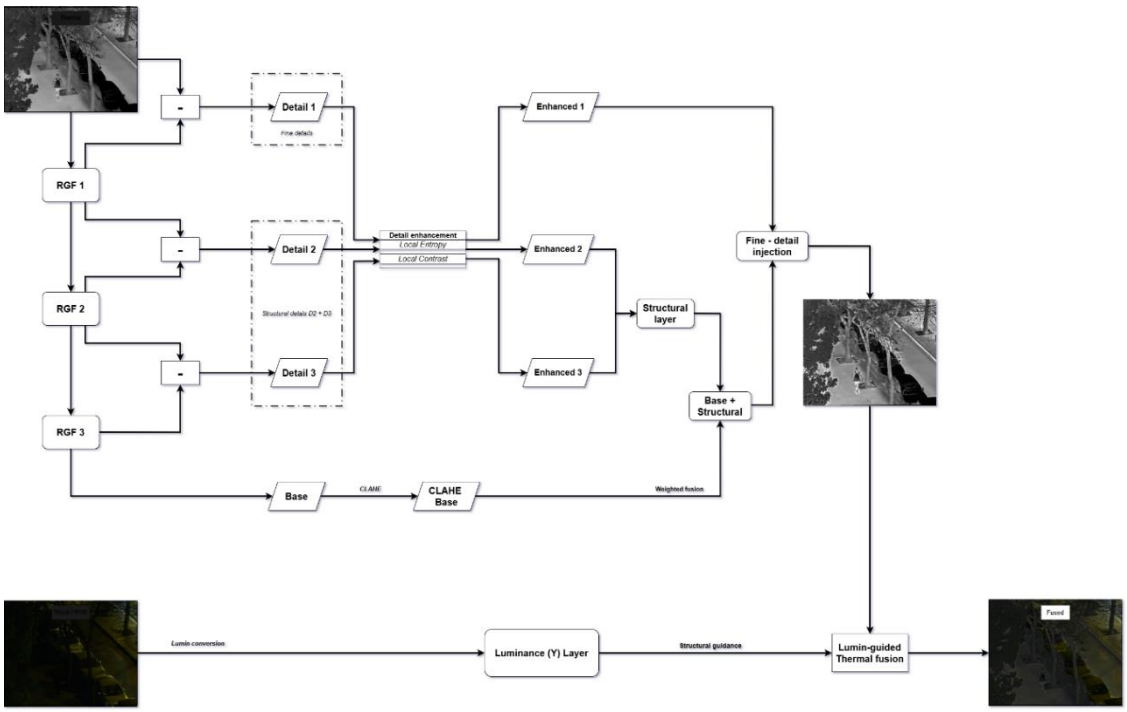


Figure 2.3: Overall architecture of the proposed method

Let T denote the input thermal image and V denote the corresponding RGB image. Both images are assumed to be captured simultaneously using a dual-sensor system and are already spatially aligned and temporally synchronized. Image registration is therefore not considered, and the focus is placed on enhancement and integration.

2.2.2. Phase 1: Thermal Image Enhancement

Based on the analysis in Chapter 1, the visual quality of thermal images is primarily degraded by three factors: limited contrast range, relatively high noise levels, and insufficient structural detail representation, particularly under challenging observation conditions such as low-light or complex environments. Therefore, thermal image enhancement must address not only global contrast improvement, but also structural preservation, selective detail enhancement, and noise control.

The proposed module is designed based on three key principles:

- Clear separation between the base layer and detail layers across multiple spatial scales, enabling isolation of different information components.
- Selective enhancement of details based on local image characteristics, instead of uniform amplification.
- Controlled enhancement strength to maintain visual stability and prevent noise amplification or artifacts.

Based on these principles, the enhancement process follows a decomposition–adjustment–reconstruction framework, allowing independent manipulation of different image components and improving overall controllability.

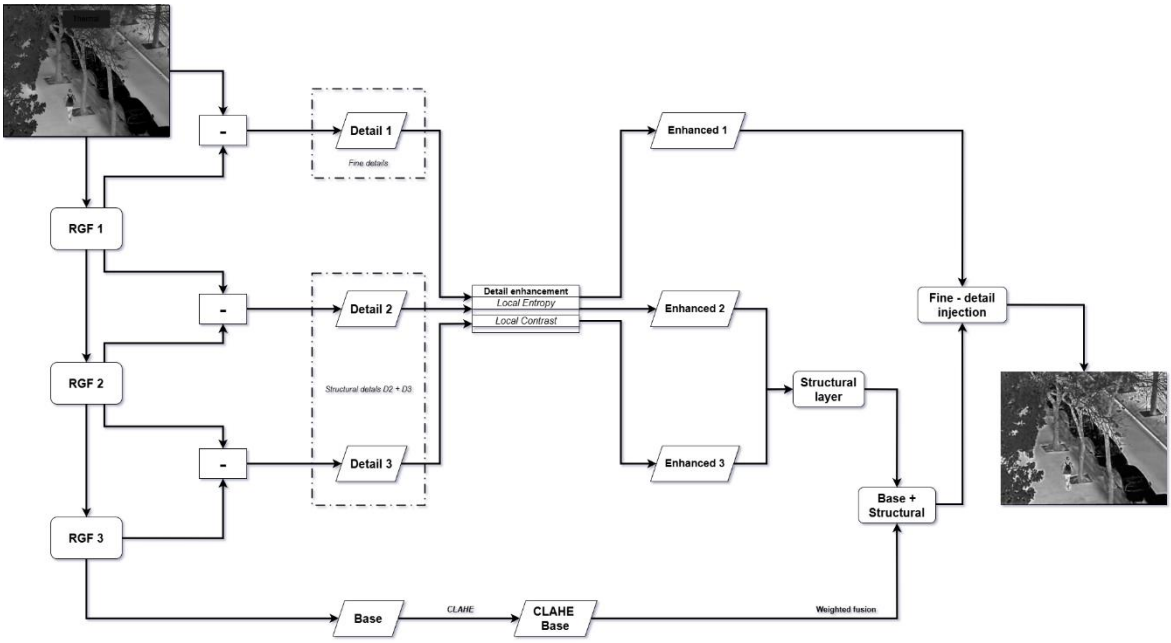


Figure 2.4: Thermal image enhancement module diagram

a. Multi-Scale Decomposition using Rolling Guidance Filter

A key challenge in thermal image enhancement is the non-uniform spatial structure of the image. Within a single thermal frame, smooth background regions, strong object edges, fine structural details, and sensor noise coexist at different spatial scales. Applying global enhancement directly often leads to undesirable effects such as noise amplification or loss of weak structural details.

To address this issue, a multi-scale decomposition strategy is adopted to separate image components prior to enhancement. The image is represented as the sum of a base layer and multiple detail layers corresponding to different spatial scales. This decomposition enables independent processing of each component and improves control over the enhancement process.

The decomposition is performed using the Rolling Guidance Filter (RGF) [1], an edge-preserving filtering algorithm based on iterative guidance. At each iteration, the output image is used as the guidance image for the next filtering step, allowing progressive removal of small-scale details while preserving major structural edges.

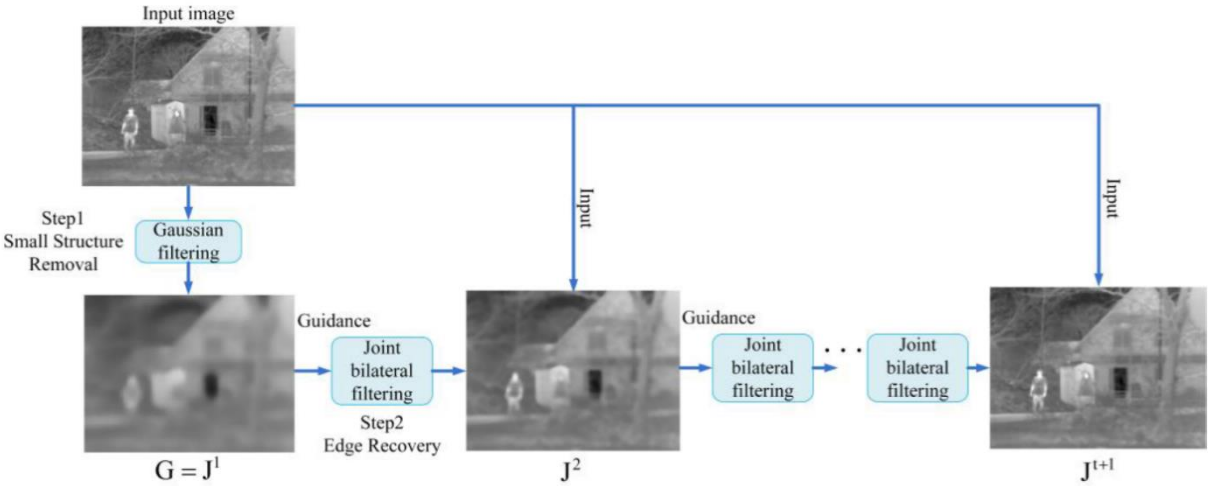


Figure 2.5: Working principle of RGF [1]

Mathematically, at iteration k, the filtered output $q_k(x, y)$ is modeled using a local linear relationship:

$$\boldsymbol{q_k(x,y) = a_k(x,y)I(x,y) + b_k(x,y)}$$

where:

- $a_k = \frac{cov(I,p)}{var(I)+\varepsilon}$, $b_k = \bar{p} - a_k\bar{I}$ with the coefficients are estimated within a local window of radius r
- p: Input image to be filtered
- ε: Regularization parameter for numerical stability

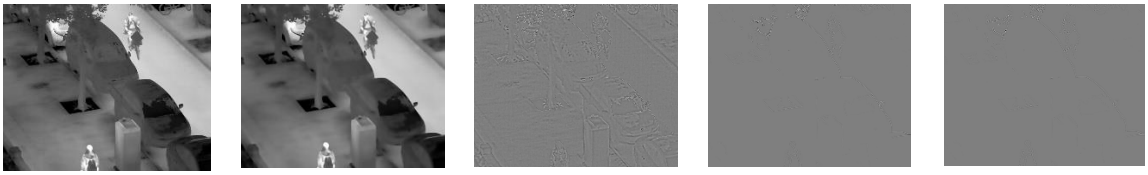
By applying RGF with different smoothing parameters, the input thermal image I is decomposed into multiple detail levels. As illustrated in Figure 2.5, the decomposition produces three detail layers and one base layer:

$$\boldsymbol{D_1 = I - P_1}$$
$$\boldsymbol{D_2 = P_1 - P_2}$$
$$\boldsymbol{D_3 = P_2 - P_3}$$

where:

- D_1: Fine detail layer
- D_2: Medium-scale detail layer

- D_3: Large-scale structural layer
- P_3: Base layer



(a) (b) (c) (d) (e)

Table 2: Multi-scale decomposition results: (a) Thermal input, (b) Base layer, (c) Large-scale structural details, (d) Medium-scale details, (e) Fine details

From an image processing perspective, this decomposition provides two key advantages. First, important structural details can be enhanced independently without affecting homogeneous background regions. Second, enhancement can be selectively applied to different detail types, reducing the risk of noise amplification. Therefore, multi-scale decomposition serves as a critical preprocessing step that enables controlled and effective thermal image enhancement.

b. Selective Detail Enhancement Based on Local Entropy and Contrast

After multi-scale decomposition, the extracted detail layers contain both meaningful structural information and noise components. Uniform enhancement across these layers may lead to excessive amplification of noise or already high-contrast regions, resulting in artifacts and visual instability.

To address this issue, a selective detail enhancement strategy is proposed. The enhancement strength at each pixel is adaptively controlled based on local characteristics of the original thermal image. Two features are employed: local entropy and local contrast.

To address this issue, a selective detail enhancement strategy is proposed. The enhancement strength at each pixel is adaptively controlled based on local characteristics of the original thermal image. Two features are employed: local

entropy and local contrast. This adaptive mechanism enables controlled enhancement of detail layers, improving structural clarity without introducing instability.

c. Base Layer Enhancement and Image Reconstruction using CLAHE

In addition to detail enhancement, the base layer must also be adjusted to improve overall image contrast. In this study, the base layer is enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). CLAHE is a variant of histogram equalization in which the image is divided into small regions, and local histogram equalization is applied to each region. The amplification of contrast is controlled with a clip limit parameter to prevent excessive noise enhancement.

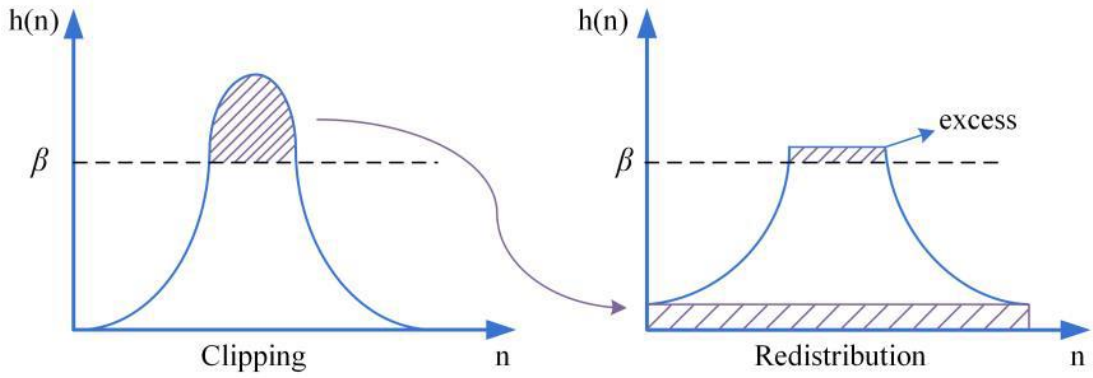


Figure 2.6: Working principle of the CLAHE method

After enhancing the base layer and adjusting the detail layers, the final enhanced thermal image is reconstructed as:

$$\boldsymbol{T_e(x,y) = B_c(x,y) + \alpha D_s(x,y) + \beta D_f(x,y)}$$

where:

- $B_c(x,y)$: Enhanced base layer
- $D_s(x,y)$: Enhanced structural detail layer
- $D_f(x,y)$: Enhanced fine detail layer

This reconstruction follows the principle of separating and independently processing image components before recombination. As a result, the output image achieves a balance between structural sharpness, contrast, and visual stability.

The algorithm can be represented in the form of pseudocode.

Algorithm 2.1: Proposed Thermal Image Enhancement Pipeline

Input: Thermal image T

Output: Enhanced Thermal image T_e

Step 1: Multi-scale decomposition

- Apply RGF with different parameters to obtain:

  B (base layer), D_f (fine detail layer), D_s (structural detail layer)

Step 2: Local feature computation

- Compute local entropy H(x,y)
- Compute local contrast C(x,y)

Step 3: Weight map construction

- Generate weight maps:

  w_fine(x,y), w_struct(x,y)

Step 4: Selective detail enhancement

- D_f = w_fine $\odot$ D_f
- D_s' = w_struct $\odot$ D_s

Step 5: Base layer enhancement

- B_c = CLAHE(B)

Step 6: Image reconstruction

- $T_e = B_c + \alpha D_s' + \beta D_f'$

Return T_e

Table 3: Pseuco-code representation of the algorithm

The proposed thermal image enhancement module follows a decomposition–adaptive enhancement–reconstruction framework. This design enables simultaneous improvement of contrast, structural sharpness, and visual stability, providing a robust foundation for subsequent multi-sensor image fusion.

The enhanced thermal image not only serves as the output of the enhancement process but also provides valuable structural information for further

integration. Therefore, the next section presents the luminance-based fusion method for combining enhanced thermal images with RGB data in multi-sensor observation systems.

2.2.3. Phase 2: Luminance-based Fusion

After thermal image enhancement, the output image exhibits improved contrast and structural representation. In practical observation systems, particularly multi-sensor systems, thermal images are often used together with visible (RGB) images to exploit complementary information from both modalities [3].

In this study, fusion is not the primary objective but is presented as an extension to demonstrate how the enhanced thermal image can support visual observation. Thermal images provide temperature-based structural information and highlight salient targets, while RGB images provide texture and color information. Therefore, a luminance-based fusion strategy is adopted to combine the advantages of both sources while preserving the natural appearance of the RGB image.

From a perceptual perspective, human vision primarily relies on luminance for interpreting edges, shapes, and spatial structures, while chrominance contributes mainly to visual appearance. Compared to direct fusion in the RGB domain, luminance-based fusion offers two advantages:

- It enables controlled integration of structural information from thermal imagery through the luminance channel.
- It enables controlled integration of structural information from thermal imagery through the luminance channel.

With that, choosing luminance is the most suitable for the proposed method and for the pipeline overall.

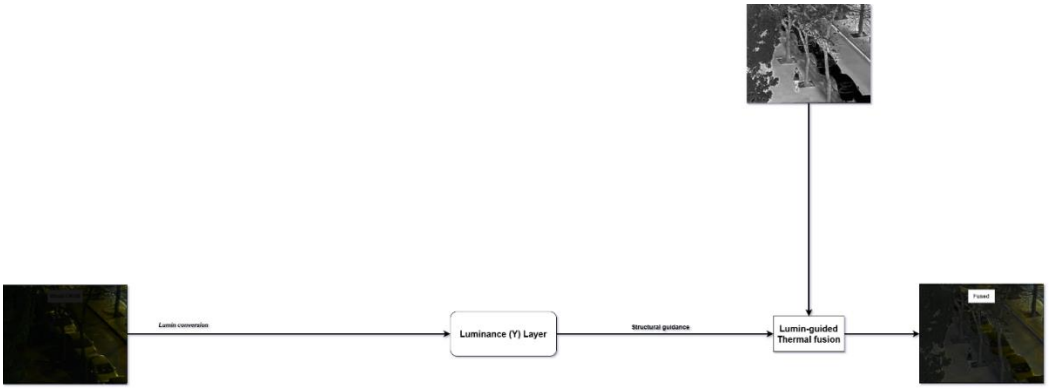


Figure 2.7: Thermal–visible image fusion based on luminance

Let $T_e(x, y)$ denote the enhanced thermal image and $V(x, y)$ denote the input RGB image. The RGB image is first transformed into the YCrCb color space:

$$\boldsymbol{Y_v(x,y), C_r(x,y), C_b(x,y) \ = \ C_{RGB \to YCrCb}(V(x,y))}$$

where:

- $Y_v(x, y)$: Luminance component of the RGB image
- $C_r(x, y), C_b(x, y)$: Chrominance components

In parallel, a thermal luminance component is constructed from the enhanced thermal image. Let $B_c(x, y)$ denote the enhanced base layer, $D_s(x, y)$ the structural detail layer, and $D_f(x, y)$ the fine detail layer. The thermal luminance is defined as:

$$\boldsymbol{Y_T(x,y) = clip(\beta_c B_c(x,y) + \beta_s D_s(x,y) + \beta_f D_f(x,y), 0, 1)}$$

where:

- $\beta_c, \beta_s, \beta_f$: Weighting coefficients for each component
- $clip(\cdot)$: Operator ensuring values remain within the valid range

If the thermal and RGB images have different resolutions, $Y_T(x, y)$ is resized to match the RGB image before fusion. The fusion is performed through controlled linear blending of luminance components:

$$\boldsymbol{Y_{out}(x,y) = \alpha Y_v(x,y) + (1 - \alpha) Y_T(x,y)}$$

where:

- $\alpha \in [0,1]$: Blending factor.
- Larger α: Output closer to original RGB luminance.
- Smaller α: Thermal structural information is emphasized.

The final fused image is reconstructed by replacing the luminance channel and transforming back to RGB:

$$\boldsymbol{V_f(x,y) = C_{YCrCb\rightarrow RGB}(Y_{out}(x,y), C_r(x,y), C_b(x,y))}$$

Since chrominance channels remain unchanged, the output image preserves natural color appearance while incorporating enhanced structural information from thermal imagery. From an implementation perspective, this approach is computationally efficient, involving only color space transformation, interpolation, and linear blending. Therefore, it is suitable for near real-time systems and resource-constrained platforms.

(a) (b)

Figure 2.8: Input visible image and fused RGB–thermal result based on the Y channel: (a) Input image, (b) Fused result

In summary, the luminance-based fusion module demonstrates that the enhanced thermal image can serve not only as a standalone output but also as a valuable structural source in multi-sensor observation systems. However, this component is presented as an extension and is not the primary focus of the study.

2.2.4. Overall evaluation

In summary, the proposed method establishes a thermal image enhancement pipeline based on multi-scale decomposition and selective detail enhancement, with an optional extension for luminance-based image fusion. The components of the method are designed to operate independently while remaining tightly integrated, enabling effective control over the enhancement process. The proposed approach simultaneously improves contrast, structural sharpness, and visual stability, addressing key limitations of existing methods while maintaining computational efficiency suitable for practical deployment

In the next chapter, experimental evaluations are conducted to quantitatively and qualitatively assess the performance of the proposed method, and to compare it with related approaches.

CHAPTER 3: SYSTEM DESIGN, IMPLEMENTATION, AND EVALUATION

3.1. System design

Based on the proposed method presented in Chapter 2, the image processing system is designed following a pipeline architecture, where input data is processed sequentially through functional modules to produce an output image with improved contrast and structural representation

3.1.1. Overall architecture

The system design aims not only to implement the proposed algorithm but also to ensure practical deployability. This includes clear modular organization, scalability, and computational efficiency. Therefore, the system is structured into independent functional blocks that directly correspond to the components of the proposed method.

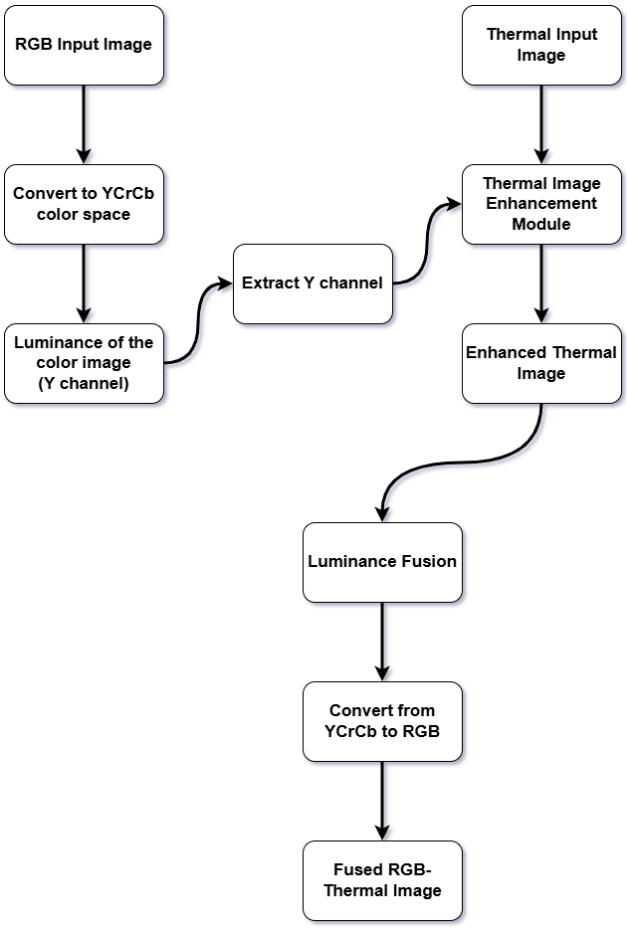


Figure 3.1: Pipeline block diagram of system modules

As shown in Figure 3.1, the system takes two input data sources: a thermal image T(x, y) and a corresponding RGB image V(x, y). These two images are assumed to be temporally synchronized and spatially aligned [3]; therefore, the system does not include an image registration step. This allows the processing to focus entirely on enhancement and information extraction.

The system processing pipeline consists of two main modules:

(1). *Thermal Image Enhancement Module*: Here it performs a sequence of operations to improve the quality of the input thermal image. The image is first decomposed into multiple scales using the Rolling Guidance Filter to separate the base layer and detail layers. The detail layers are then selectively enhanced based on local features, while the base layer is refined using the CLAHE method. The output of this module is an enhanced thermal image with significantly improved contrast and structural representation.

(2). *Thermal–RGB Fusion Module*: Here it utilizes the enhanced thermal image to support observation in multi-sensor systems. The fusion process is performed in the luminance domain, where structural information from the thermal image is integrated into the luminance channel of the RGB image, while the chrominance components are preserved. This approach enhances structural perception while maintaining the natural color appearance of the RGB image.

The pipeline architecture with independent modules provides several important advantages:

- Clear control over each processing step.
- Ease of evaluation and optimization for individual modules.
- Suitability for deployment on systems with limited computational resources.

This organization ensures a one-to-one correspondence between the theoretical model and the system implementation, thereby reducing discrepancies during development. It also provides a foundation for future extensibility,

allowing individual modules to be replaced or improved without affecting the overall pipeline.

3.1.2. Design of Processing Modules

Based on the overall architecture presented in the previous section, the image processing system is implemented as a set of functional modules that directly correspond to the components of the proposed method. These modules are developed in C++ using the OpenCV library, with the objectives of ensuring computational efficiency, fine-grained algorithm control, and flexibility for practical deployment.

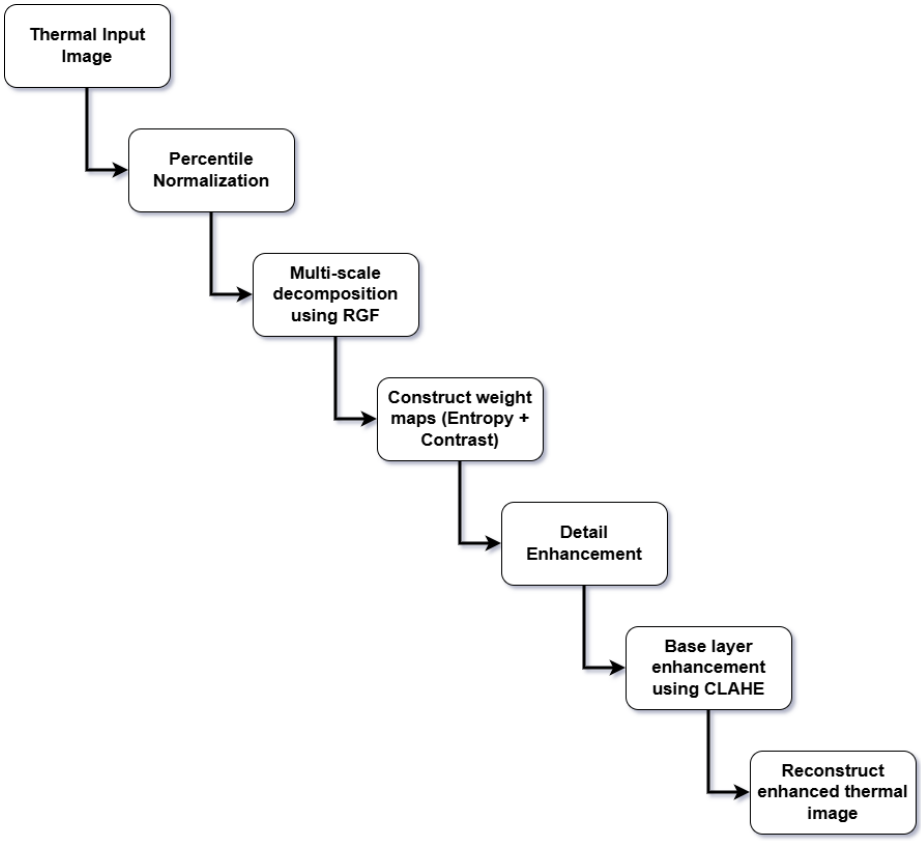


Figure 3.2: Processing flow diagram by modules

a. Thermal Image Preprocessing and Normalization Module

This module is responsible for standardizing the input data to ensure stability throughout the processing pipeline. In the system, this step is implemented through the function:

Mat percentile8bit(const Mat &img, const float p_lo, const float p_high)

In terms of processing, the input image is first checked for the number of channels and converted to grayscale if necessary. The image is then converted to floating-point format to support statistical computations. All pixel values are subsequently flattened into a one-dimensional vector for percentile calculation.

The lower and upper percentile thresholds are determined using the std::nth_element algorithm, which significantly reduces computational cost compared to full sorting. After obtaining the thresholds p_lo and p_high, the image is linearly mapped to the range [0, 255], ensuring effective utilization of the dynamic range. Compared to simple normalization methods, percentile-based normalization offers several advantages:

- Reduced sensitivity to outliers.
- Improved effective contrast.
- Better input quality for subsequent local processing steps.

As a result, this module serves as a necessary preprocessing stage, improving system robustness when handling thermal images with non-uniform intensity distributions.

b. Multi-Scale Decomposition Module based on RGF

This module is the core component of the system, responsible for decomposing the image into components at multiple spatial scales. It is implemented through the functions *rolling_guided_filter*, *rgf_multi_scale*, *rgf_decompose* respectively.

The function *rolling_guided_filter* performs iterative filtering, where the guidance image is updated after each iteration. At each step, guided filtering is applied using a local linear model:

$$\boldsymbol{q}(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{a}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{I}(\boldsymbol{x},\boldsymbol{y}) + \boldsymbol{b}(\boldsymbol{x},\boldsymbol{y})$$

This iterative mechanism progressively removes small-scale details and noise while preserving major structural edges. This property is particularly

important for thermal images, where objects often exhibit low contrast relative to the background.

After filtering with different parameter settings, the image is decomposed into multiple detail levels:

(a) (b) (c)

Figure 3.3: Multi-scale decomposition results:
(a) Large details, (b) Medium details, (c) Fine details

The purpose of this module is to provide a structured representation of image components, enabling subsequent enhancement steps to operate on each type of information independently rather than uniformly across the entire image. This improves the system's ability to balance detail enhancement and noise suppression.

c. Weight Map Construction Module

After image decomposition, the system determines the appropriate enhancement strength for each image region. This module is implemented through the functions *local_entropy, local_contrast, weight_map.*

Weight maps are constructed to regulate the enhancement level for each detail layer, preventing excessive amplification in noisy or already high-contrast regions while maintaining overall visual stability.

(1). Local Entropy: it is computed within sliding windows to reflect the information richness at each pixel location. To improve performance, a lookup table (LUT) is employed, avoiding repeated logarithmic computations during processing.

(2). Local Contrast: it is estimated using variance and standard deviation within local neighborhoods. In the implementation, the boxFilter function is used to accelerate computation, ensuring efficiency even for large images.

Figure 3.4: Enhancement process: Input image, Entropy map, Contrast map, Enhanced image

This module plays a critical intermediate role, as it determines how strongly the detail enhancement step is applied at each region, enabling the system to shift from global processing to content-adaptive processing.

d. Detail Enhancement Module

This module performs enhancement of the detail layers using the weight maps constructed in the previous step. In the implementation, this process is realized through the components *d_fine_enhanced* và *d_struct_enhanced*. Specifically, the fine detail layer is modulated using the corresponding weight map and constrained within a predefined range to prevent over-enhancement. This constraint is essential because fine details are more susceptible to being confused with noise. In contrast, the structural detail layer is enhanced more aggressively to emphasize object boundaries and dominant geometric structures within the scene

This design enables the system to distinguish between two types of information:

- Fine details, which should be preserved but carefully controlled.
- Structural details, which can be more strongly enhanced to improve observability.

As a result, the output image achieves improved structural sharpness while maintaining visual stability and avoiding artifacts or excessive noise amplification.

e. Base layer Enhancement Module (CLAHE)

In parallel with detail enhancement, the base layer is adjusted to improve overall image contrast. In the system, this is implemented using the CLAHE module, corresponding to the function clahe.

In principle, CLAHE is a variant of traditional histogram equalization. Instead of operating on the entire image, CLAHE divides the image into smaller regions and applies local histogram equalization to each region. The contrast amplification is controlled with a clip limit parameter, preventing excessive noise amplification in low-variance areas.

Applying CLAHE to the base layer enhances global contrast, improves visibility in low-intensity regions, and clarifies subtle structures without compromising image stability.

This module plays a balancing role before image reconstruction. It ensures that the final output image not only contains enhanced details but also maintains an appropriate global contrast level for visual interpretation.

f. Thermal image Reconstruction Module

After the base layer and detail layers have been processed independently, the system performs reconstruction to produce the final enhanced thermal image. This module corresponds to the function *build_luminance*. The enhanced image is reconstructed as::

$$\boldsymbol{T_e = B_c + \beta_s D_s + \beta_f D_f}$$

In this step, the CLAHE-enhanced base layer and the selectively enhanced detail layers are combined using a weighted mechanism. The base layer maintains the overall intensity stability of the image, while the detail layers enhance structural information and emphasize important regions.

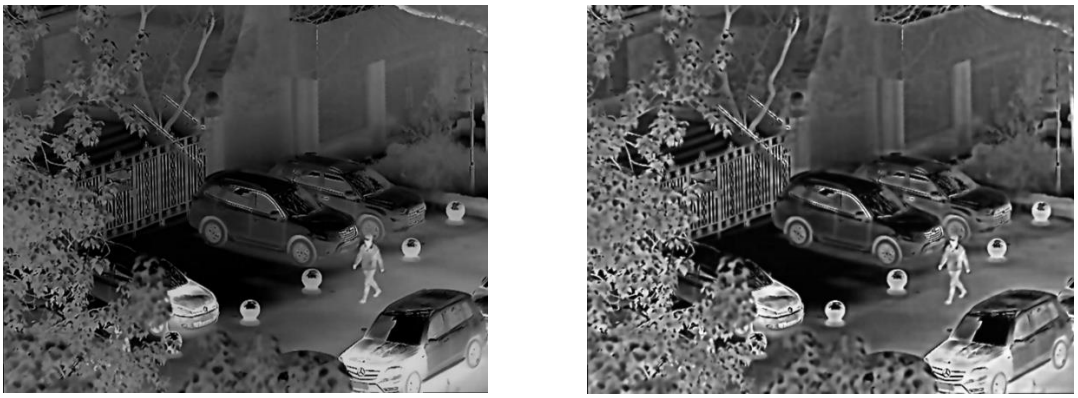


Figure 3.5: Thermal image before and after enhancement

Notably, before reconstruction, the fine detail layer is further constrained to prevent over-enhancement in noise-prone regions. The components are then combined and normalized to the valid intensity range.

This design allows the system to effectively balance structural sharpness, overall contrast, and visual stability. As a result, the reconstructed image not only improves detail representation but also preserves a natural balance between background and objects, consistent with the goal of enhancing thermal image observability.

g. Image fusion Module

The final module of the system performs fusion between the enhanced thermal image and the RGB image, implemented through the function *luminance_fusion*. This step extends the pipeline by incorporating color information from the RGB image while maintaining the central role of the enhanced thermal data.

In the system, the input RGB image is first converted to the YCrCb color space. The luminance channel Y is extracted for the fusion process, while the enhanced thermal image is treated as an additional luminance component. These two components are then combined through controlled linear blending to produce a new luminance channel.

After computing the fused luminance, it is reintegrated into the YCrCb image, followed by conversion back to the RGB color space. The resulting image demonstrates improved structural representation and object visibility due to the thermal information, while preserving the natural color characteristics of the original RGB image.

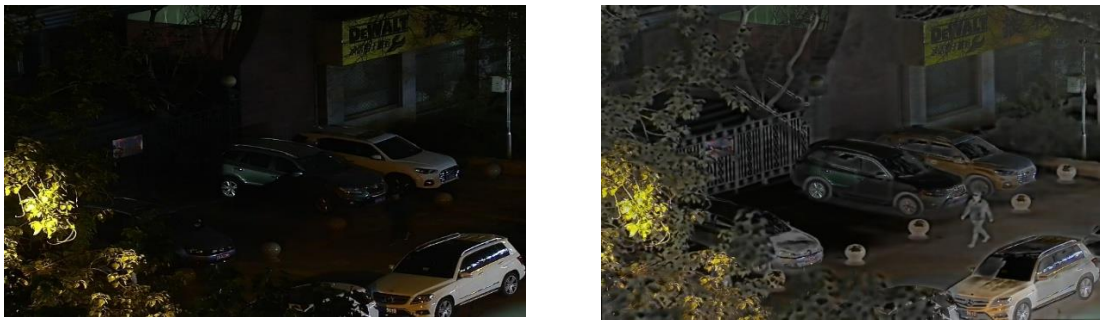


Figure 3.6: Input visible image and fused result

Within the scope of this study, this module is not the primary focus but is presented as an extension of the enhanced thermal image. Nevertheless, it demonstrates that the enhanced thermal output can be effectively integrated into multi-sensor observation systems.

Overall, the system is designed with a clear modular structure, where each module performs a distinct function while remaining closely integrated with the others. This organization facilitates independent evaluation and optimization of each component, and supports efficient system development. As a result, the system not only satisfies academic requirements but also demonstrates practical applicability and scalability for future deployment and extension.

3.1.3. Pipeline organization & Data flow

Based on the overall architecture and processing modules presented in the previous sections, the system is organized as a sequential pipeline, where data flows through functional modules in a predefined order. This organization clarifies the processing flow, while facilitating control, evaluation, and optimization of individual components. At the same time, the pipeline structure

ensures consistency between the proposed method in Chapter 2 and its implementation.

The processing flow begins with two input sources: the thermal image *T(x,y)* and the RGB image *V(x,y)*.

These inputs are initially processed through two separate branches, corresponding to two objectives: thermal image enhancement and color information extraction for fusion. This separation maintains modular independence and allows individual components to be modified or replaced without affecting the overall pipeline.

For the thermal image branch, processing is performed through a sequence of steps designed to balance contrast, structural detail representation, and visual stability. First, the input thermal image is normalized using percentile-based scaling to reduce the influence of outliers and effectively utilize the dynamic range. The normalized image is then passed to the multi-scale decomposition module based on the Rolling Guidance Filter, where it is separated into a base layer and multiple detail layers at different spatial scales. This decomposition forms the foundation for subsequent processing by enabling independent handling of different image components.

Next, weight maps are constructed based on local image features, specifically local entropy and local contrast. These maps represent the relative importance of information at each location and are used to guide adaptive detail enhancement. Based on the computed weights, the detail layers are selectively enhanced: fine details are carefully controlled to avoid noise amplification, while structural details are emphasized to improve object visibility.

In parallel, the base layer is moderately enhanced using CLAHE to improve overall contrast without introducing visual instability. After these independent processing steps, the base and detail components are recombined to produce the enhanced thermal image $T_e(x, y)$, which serves as the primary output of this branch.

For the RGB image branch, processing is simplified to ensure computational efficiency. The input RGB image is converted to the YCrCb color space to separate luminance and chrominance components. The luminance channel $Y_v(x, y)$ is extracted for fusion, while the chrominance channels are preserved to maintain color fidelity.

After both branches are processed, the system performs the fusion stage. The luminance component derived from the enhanced thermal image is resized to match the resolution of the RGB image if necessary, ensuring spatial alignment. The luminance channels are then combined via controlled linear blending, allowing adjustment of the relative contribution from each source. The resulting luminance is reinserted into the YCrCb image and converted back to RGB to produce the final output image.

The entire pipeline is implemented within the central function rgb_fusion, where modules are executed sequentially according to the described processing flow. This organization ensures a close correspondence between algorithm design and implementation, while enabling effective control of data flow and performance evaluation at each processing stage.

3.2. Implementation and Technologies

3.2.1. Programming Language and Libraries

The system is implemented in C++ using the OpenCV library. C++ is selected over high-level languages such as Python to ensure computational efficiency and suitability for deployment in time-constrained observation systems.

a. C++ Programming Language

C++ is used as the primary implementation language for the entire processing pipeline. Compared to high-level languages, C++ provides direct control over memory and data structures with efficient CPU utilization and integrates easily with hardware and embedded systems.

And, for high-resolution image processing and near real-time applications, C++ significantly reduces execution time and latency.

b. OpenCV Library

OpenCV is used as the core image processing library. The following components are utilized:

- Basic operations: color conversion, resizing, normalization…
- Filtering: boxFilter, Gaussian blur…
- Guided Filter: module ximgproc
- CLAHE: local contrast enhancement
- Matrix operations: *cv::Mat*

The library provides optimized implementations, reducing development time while maintaining high performance.

c. Initial Development in Python

The algorithm was initially prototyped and validated in Python (PyCharm environment). This stage enabled for rapid prototyping, efficient parameter tuning and comparative evaluation between methods.

After validation, the pipeline was reimplemented in C++ to meet performance and deployment requirements.

3.2.2. Development Environment

The system is developed using a C++ environment with support for modern standards (C++17 or later). CLion is used as the primary IDE, providing tools like CMake Integration, debugging tools and code analysis support. The IDE serves only as a development tool and does not affect algorithm design or runtime performance.

3.2.3. Graphical User Interface

A graphical user interface (GUI) is implemented using the Qt framework to support visualization, testing, and analysis.

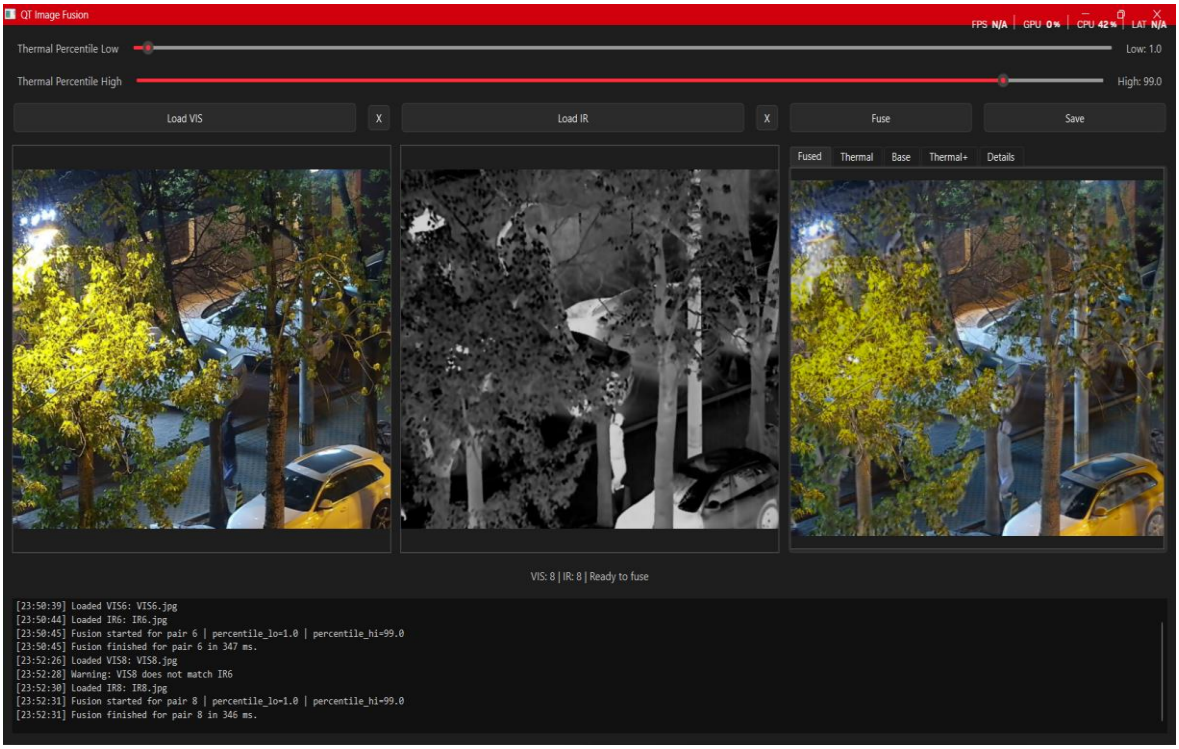


Figure 3.7: GUI implemented using Qt

The interface is connected to the core processing function *rgb_fusion* via the Qt signal–slot mechanism. The GUI provides a unified visualization of input, intermediate and output images. It also consists of a direct interaction with system parameters and real-time execution of the processing pipeline.

The interface is connected to the core processing function rgb_fusion via the Qt signal–slot mechanism:

- Visual Image: Left
- Thermal Image: Center
- Fused Image: Right

This layout reflects the processing pipeline and enables direct visual comparison between inputs and outputs.

a. Parameter Control

The parameter panel provides controls for *Percentile Low* and *Percentile High*. These parameters control thermal image normalization and allow adjustment of the input intensity range. This is critical for handling thermal

images with unstable intensity distributions and improving downstream processing performance.

b. Functional Controls

The main control buttons include:

- Load VIS: load RGB images
- Load IR: load thermal images
- Fuse: execute the pipeline process
- Save: export the output image

The control flow follows the processing sequence: load → process → save.

c. Result Visualization

The result panel provides multiple tabs for inspection.

- Fused: final fused output
- Thermal: enhanced thermal image
- Base: CLAHE-enhanced base layer
- Details: detail components

This structure enables step-by-step verification and debugging of the pipeline.

d. Logging and Performance

The bottom panel displays the processing status and the execution time per stage. This information is used for performance analysis and evaluation of near real-time capability.

e. System Role of GUI

The GUI functions as an analysis and control interface rather than a core processing component.

It is responsible for handling user input, invoking the processing pipeline through the rgb_fusion function, and displaying intermediate and final results. The interface does not participate in algorithm execution or modify processing

logic. This separation ensures that changes to the GUI do not affect computational behavior, while allowing the processing pipeline to be integrated into other environments without dependency on the interface.

3.3. System functionality

The system is developed with the primary objective of enhancing thermal image quality and supporting observation in multi-sensor environments. From an implementation perspective, the system not only executes the image processing pipeline but also provides a set of functions that enable user interaction, testing, and visual evaluation of processing results.

Accordingly, the system functionality is organized into three main groups, including image processing, data input and management, and evaluation support.

3.3.1. Image Processing Function

The core function of the system is the execution of the thermal image enhancement and image fusion pipeline based on the proposed method. The entire processing workflow is encapsulated within a central module, ensuring both automation and consistency in operation.

Once the input data is provided, the system performs a sequence of processing steps, starting from preprocessing and normalization, followed by multi-scale decomposition, selective detail enhancement, base layer enhancement, and image reconstruction. Depending on the operating mode, the final output may be either an enhanced thermal image or a fused RGB–thermal image.

A key design characteristic is that the pipeline operates fully automatically without requiring manual intervention during execution. This ensures stability and reproducibility of results, particularly when processing large datasets.

In addition to the final output, the system also retains intermediate results, including detail layers, the base layer, and weight maps. These intermediate outputs serve not only for visualization purposes but also provide insight into the internal behavior of the algorithm during processing.

Figure 3.8: Pipeline results (RGB input + Thermal input + Fused output)

3.3.2. Data Input and Management Function

The system is designed to support flexible input and management of image data in order to accommodate a variety of experimental scenarios. Users provide two primary types of input data, namely thermal images and corresponding RGB images, through the graphical interface. Common image formats are supported to ensure compatibility and ease of use.

After the data is loaded, an automatic validation and standardization process is performed to ensure input consistency. The system verifies image properties such as the number of channels, dimensions, and data format, and applies necessary transformations, including grayscale conversion for thermal images and data type normalization for subsequent processing.

In cases where the input images differ in resolution, the system automatically performs interpolation to ensure spatial correspondence between the two data sources. This step is required to support accurate fusion in later stages of the pipeline. The data input and management process is fully automated, reducing manual operations and ensuring consistency throughout the processing pipeline. This is particularly important when the system is applied to multiple datasets, where stability and reproducibility must be maintained.

3.3.3. Parameter Adjustment Function

The system allows users to adjust several key parameters that directly influence processing quality through the graphical interface.

Typical parameters include:

- Percentile thresholds (p_low, p_high)
- Luminance blending coefficient (α)
- Parameters related to kernel size or local processing windows

These parameters control different stages of the pipeline, including normalization, detail enhancement, and image fusion. These parameters control different stages of the pipeline, including normalization, detail enhancement, and image fusion.

Notably, parameter updates are applied in near real-time, allowing users to directly observe the impact of each parameter on the output image. This capability is particularly useful during experimentation and algorithm optimization.

3.3.4. Visualization and Analysis Function

A key strength of the system lies in its ability to provide comprehensive visual representation of the processing pipeline. The interface is designed to display multiple outputs simultaneously, including:

- Input Visual Image
- Input Thermal Image
- Enhanced Thermal Image
- Final Fused Output

In addition to these primary outputs, the system also provides access to intermediate results through dedicated tabs. These include:

- Base layer
- Different scale detail layers (D1, D2, D3)
- Weight maps (Entropy, Contrast)

The availability of intermediate visualization allows users to better understand the internal behavior of the algorithm and how each processing stage

contributes to the final result. It also facilitates identification of potential issues, such as noise amplification or loss of structural details, during the processing pipeline.

Figure 3.9: GUI with tabs (Fused / Thermal / Base / Details)

3.3.5. Result Storage Function

In addition to processing and visualization, the system provides functionality for saving output images to support analysis, comparison, and reporting. After processing is completed, users can conveniently store the results through the graphical interface.

The storage functionality is not limited to the final fused output but also includes intermediate results within the pipeline. These may consist of the enhanced thermal image, the CLAHE-processed base layer, and the processed detail layers. Retaining these components supports deeper analysis of algorithm behavior, particularly in cases requiring quantitative evaluation or comparison with other methods.

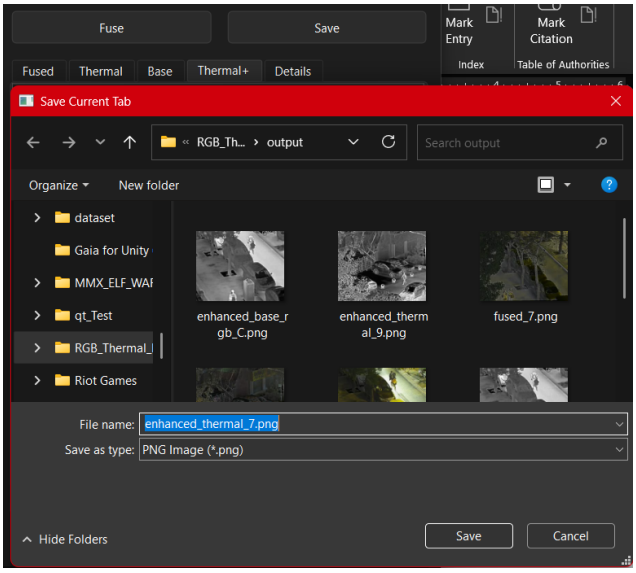


Figure 3.10: Save button and output file

From a research perspective, the storage function plays an important role in constructing a dataset of processed results for subsequent experiments and evaluation. It also ensures that results can be reused in extended studies or further algorithm refinement.

3.4. Experiments and Evaluation

3.4.1. Evaluation Objectives and Methodology

The objective of the experimental process is to provide a comprehensive evaluation of the proposed thermal image enhancement method in terms of visual quality, structural preservation, and computational cost. The evaluation is conducted along two complementary directions, including quantitative assessment using representative metrics and qualitative assessment based on visual inspection.

Within the scope of this study, the selected quantitative measures are intended to reflect key characteristics of the enhanced images, including:

- Level of information content
- Overall contrast
- Structural sharpness

- Edge intensity related to object visibility

In addition, computational performance is considered as an important criterion, particularly in the context of potential deployment in practical observation systems.

The evaluation is performed on three methods, including:

- Rolling Guidance Filter (RGF)
- Multi-Scale Guided Filter (MSGF)
- Proposed method (Hybrid)

This comparison is designed to position the proposed method relative to representative traditional approaches for thermal image enhancement.

3.4.2. Dataset and Experimental Setup

The experiments are conducted on paired thermal–visible images from the LLVIP dataset, which consists of synchronized image pairs captured using two sensors. The dataset includes a variety of observation conditions, such as low-light environments, complex backgrounds, and multiple objects within the scene. The input images are assumed to be temporally synchronized and spatially aligned; therefore, the experiments focus exclusively on image enhancement and fusion without considering image registration.

The proposed method is implemented in C++ using the OpenCV library. For consistency, the comparison methods are also implemented within the same environment to ensure a fair evaluation.

The parameters of the proposed method are selected based on empirical testing to achieve a balance between image quality and computational efficiency. Specifically:

- Percentile normalization range: [1%, 99%].
- Window sizes for entropy and contrast computation according to image resolution.

- Weighting coefficients β and α are adjusted to avoid over-enhancement while maintaining stable visual output.

3.4.3. Evaluation metrics

All metrics are computed on 8-bit grayscale images I_8, normalized to the range [0, 255], to ensure consistency across all evaluated methods. The use of multiple metrics allows the results to be examined from complementary perspectives.

Since no single metric can fully represent image quality, the evaluation relies on a combination of measures to capture information content, contrast, structural detail, and edge strength.

a. Image Entropy

Entropy is used to quantify the distribution of intensity values in the image. In this study, it serves as an indicator of how much information is present after enhancement. Higher entropy generally corresponds to a broader intensity distribution; however, it may also increase in the presence of noise. For this reason, entropy is considered alongside other metrics rather than in isolation..

b. RMS Contrast

RMS contrast measures the variation of intensity values across the image. It reflects the degree of separation between different regions, which is particularly relevant for thermal images where inherent contrast is often limited. In the evaluation, this metric is used to assess the effectiveness of contrast enhancement, especially in the base layer.

c. Laplacian Variance

Laplacian variance is used as an indicator of image sharpness by capturing the strength of high-frequency components. It reflects the presence of structural details and fine variations in intensity. In this study, it is used to evaluate how well detail information is preserved and enhanced after processing.

d. Sobel Energy

Sobel energy measures the overall strength of edges detected in the image. It provides an estimate of how clearly structural boundaries are represented. Within this evaluation, it is used to assess the visibility of object contours and the effectiveness of detail enhancement.

3.4.4. Result analysis

Method	Entropy	RMS	LapVar	Sobel	Time (ms)
RGF	7.246	47.37	10.26	1711.63	378.56
MSGF	7.571	53.74	38.19	3665.69	481.25
Proposed method	7.456	52.05	50.58	4472.31	380.20

Table 4: Comparison of evaluation metrics across methods over 1000 images

From the computational perspective, the proposed method achieves a processing time comparable to RGF and lower than MSGF. This indicates that the method maintains a similar level of computational efficiency while incorporating additional enhancement steps.

a. Entropy results

In terms of entropy, MSGF achieves the highest value, indicating a stronger redistribution of intensity levels. The proposed method yields a slightly lower entropy value but remains within a comparable range. This suggests that the method improves information content while avoiding excessive redistribution that may affect visual stability.

b. RMS results

For RMS contrast, the proposed method achieves a value close to MSGF and higher than RGF. This indicates that the method effectively improves global contrast, contributing to better separation between regions in the image.

c. Laplacian results

Regarding Laplacian variance, the proposed method obtains the highest value among the three methods. This reflects stronger preservation and enhancement of structural details, particularly in high-frequency components.

d. Sobel results

Regarding Laplacian variance, the proposed method obtains the highest value among the three methods. This reflects stronger preservation and enhancement of structural details, particularly in high-frequency components.

e. Visual judgement

From visual inspection:

- RGF produces smoother images but tends to lose fine details.
- MSGF enhances prominent structures but may suppress background details in certain regions.
- The proposed method maintains a balance between detail enhancement and visual stability, with improved representation of small structures and low-contrast objects.

Small structures and low-contrast objects are represented more clearly in the results produced by the proposed method.

Figure 3.11: Comparison between input and results of different methods.
Left to right: Input – RGF – Proposed method – MSGF

3.4.5. Overall Evaluation

Based on the quantitative and qualitative results presented in the previous sections, the proposed method achieves a well-balanced performance across the key criteria in thermal image enhancement. Unlike conventional approaches that tend to prioritize a single aspect, such as smoothing or contrast enhancement, the method in this study is designed to regulate multiple image components simultaneously, including the base layer and detail layers at different spatial scales. This design enables a more comprehensive improvement in image quality, rather than localized enhancement at the expense of other characteristics.

From the perspective of information content, the entropy results show that the proposed method does not achieve the highest value compared to MSGF, but remains at a relatively high and stable level. This reflects an intentional design choice: the method does not aim to maximize entropy, but instead maintains structural consistency and visual stability. In thermal imaging, excessive increases in entropy may lead to noise amplification or disruption of natural intensity separation. Therefore, the balanced behavior observed in the proposed method is consistent with practical requirements.

In terms of contrast, the proposed method achieves an RMS value close to that of MSGF and higher than that of RGF. This indicates that the combination of base layer enhancement using CLAHE and detail injection from multi-scale components effectively improves intensity separation. The use of adaptive weighting based on local entropy and contrast further avoids uniform enhancement, which can otherwise lead to visual imbalance.

The most notable improvement is observed in the structure-related metrics, namely Laplacian variance and Sobel energy. The higher values compared to the reference methods indicate stronger preservation and enhancement of structural details. This is particularly relevant in observation tasks, where objects are primarily identified through edges and shape. The results suggest that the multi-scale decomposition combined with selective detail enhancement is effective in reinforcing structural information.

From a computational perspective, the proposed method exhibits processing time comparable to RGF and lower than MSGF, despite involving additional processing stages. This indicates that the modular pipeline design and the use of optimized operations in OpenCV help maintain a reasonable computational cost. However, the overall processing time of the full pipeline does not yet meet strict real-time requirements, especially for high-resolution inputs. Further optimization, such as parallel processing or hardware acceleration, would be required for deployment in time-critical systems.

From visual inspection, the output images demonstrate improved observability. Objects are more clearly distinguishable, and structural details are better preserved compared to RGF, while avoiding the suppression of background details observed in MSGF. At the same time, the background remains stable, without noticeable artifacts or excessive noise. This indicates that the method achieves a balance between enhancement and visual stability, which is essential for practical observation systems.

Overall, the proposed method demonstrates consistent effectiveness in improving thermal image quality, particularly in terms of structural preservation and enhancement. Although it does not maximize individual metrics, it achieves a balanced performance across multiple evaluation criteria, aligning with the objective of improving visual observability in practical applications. This balance represents the core strength of the method and provides a foundation for further refinement and future development.

3.5. Discussions and Remarks

3.5.1. Overall Assessment of the Proposed Method

The proposed method achieves the primary objective of improving the visual quality of thermal images, particularly in terms of contrast enhancement, structural representation, and object–background discrimination. By organizing the processing pipeline around multi-scale decomposition combined with selective detail enhancement, the resulting images demonstrate improved observability, especially in regions with low contrast or subtle structures.

A notable characteristic of the method is its ability to balance detail enhancement and noise control. In many traditional approaches, enhancing detail often leads to noise amplification or instability in homogeneous regions. In this work, the explicit separation between the base layer and multiple detail layers enables independent control over each component, thereby reducing interference between structural information and noise. This contributes to a more stable and visually consistent output.

A notable characteristic of the method is its ability to balance detail enhancement and noise control. In many traditional approaches, enhancing detail often leads to noise amplification or instability in homogeneous regions. In this work, the explicit separation between the base layer and multiple detail layers enables independent control over each component, thereby reducing interference between structural information and noise. This contributes to a more stable and visually consistent output.

From an implementation perspective, the method does not rely on training data or complex learning models. This reduces computational overhead and simplifies deployment, while ensuring stable behavior across different operating conditions. The independence from training data also makes the method suitable for systems with limited data availability.

In addition to visual improvement, the enhanced images provide a more informative representation that can support subsequent image processing tasks,

such as object detection or segmentation, although these tasks are beyond the scope of the present study.

3.5.2. Comparison with Related Methods

Existing thermal image enhancement approaches can be broadly categorized into traditional methods and deep learning-based methods. Each category presents distinct advantages and limitations, and the proposed method can be positioned as a balanced solution between performance and practicality.

Traditional methods such as Rolling Guidance Filter (RGF) are effective in smoothing and noise reduction, but often suppress fine details due to their iterative filtering nature. Multi-Scale Guided Filtering (MSGF), on the other hand, enhances structures across multiple spatial scales, improving the visibility of prominent objects. However, it may also reduce less salient background details.

The proposed method integrates elements from both approaches. Multi-scale decomposition based on RGF enables effective separation of image components, while adaptive detail enhancement mitigates the limitations of uniform processing. As a result, the method achieves improved balance between detail preservation and noise control, which is difficult to obtain using either method alone.

Compared to deep learning-based approaches such as DenseFuse or U2Fusion, the proposed method does not consistently achieve the highest visual quality in all scenarios, particularly in complex scenes requiring strong nonlinear representation. However, it offers clear advantages in transparency and interpretability. Each stage of the pipeline is well-defined and can be analyzed independently, unlike deep models where internal representations are less accessible.

In addition, deep learning approaches typically require significant computational resources and training time, whereas the proposed method can be deployed on CPU-based or resource-constrained systems. This makes it more suitable for practical applications where efficiency and reliability are critical.

3.5.3. Method limitations

Despite the positive results, several limitations of the proposed method should be acknowledged.

First, the method depends on a number of parameters, including window sizes, weighting coefficients, and normalization thresholds. The choice of these parameters has a direct impact on output quality. In some cases, suboptimal parameter selection may lead to less effective enhancement, indicating that the method is not fully self-adaptive across all input conditions.

Second, the computational cost of the method, particularly in multi-scale decomposition and local feature computation, becomes significant for high-resolution images. Although several optimizations have been applied, achieving real-time performance remains challenging in certain scenarios.

Third, the method focuses primarily on enhancing thermal images and only partially exploits additional information sources. The fusion module operates mainly on the luminance component, which limits the use of richer color information available in RGB images.

Furthermore, the evaluation in this study relies on no-reference metrics and visual inspection. While appropriate for the problem setting, these approaches may not fully capture all aspects of perceived image quality in every scenario.

3.5.4. Parameter Sensitivity Analysis

Several parameters in the proposed method play a critical role in controlling the processing behavior and directly affect the output quality. Therefore, analyzing parameter sensitivity is necessary to assess system stability and adaptability.

The percentile parameters (p_{low}, p_{high}) in the normalization stage influence the dynamic range of the input image. If the selected range is too narrow, important information in bright or dark regions may be lost. Conversely, an overly wide range reduces the effectiveness of contrast enhancement. Empirical observations indicate that values around [1%, 99%] provide a reasonable balance across different thermal images.

In the multi-scale decomposition stage, parameters such as filter radius, number of iterations, and the regularization factor ε determine the degree of smoothing and separation between detail layers. Larger radii produce smoother base layers but may suppress important details, while smaller values preserve details but reduce noise removal effectiveness. Appropriate parameter selection is therefore required to balance these effects.

In the detail enhancement stage, weighting coefficients (w_{fine} and w_{struct}) control the amplification of different detail layers. Excessive weighting may lead to over-enhancement and noise amplification, whereas insufficient weighting reduces the effectiveness of enhancement. The use of adaptive weighting based on local entropy and contrast reduces sensitivity to fixed parameter values, allowing the system to adjust according to image content.

For the CLAHE module, parameters such as clip limit and tile size affect base layer contrast. High clip limits may amplify noise, while low values limit enhancement effectiveness. Moderate values provide a balance between contrast improvement and stability.

Finally, the fusion coefficient α determines the relative contribution between thermal and visible information. Larger values preserve the original appearance of the visible image, while smaller values emphasize thermal structure. Intermediate values generally provide a balanced result.

Overall, although the method depends on several parameters, the combination of adaptive mechanisms and structured pipeline design ensures relatively stable performance across a range of parameter settings.

3.5.5. Scalability and Future Work

From the results gathered above, the proposed method presents several opportunities for further development.

One important direction is performance optimization, particularly for computationally intensive stages such as multi-scale decomposition and local

feature extraction. Implementation on parallel architectures, such as GPUs or FPGAs, could significantly improve processing speed and enable real-time applications.

In addition, the set of local features used in the method can be extended or refined to improve the distinction between structural details and noise. Developing automatic parameter tuning mechanisms based on image content is another promising direction to enhance adaptability.

The fusion module can also be extended to exploit additional information from RGB images or integrate other modalities, such as depth or sensor data. This would improve representation capability in multi-sensor systems.

Finally, combining traditional processing techniques with learning-based components in a hybrid framework may offer further improvements. Such integration could leverage the strengths of both approaches while maintaining interpretability and practical deployability.

CONCLUSION

This study proposes a method for enhancing thermal images based on traditional image processing techniques. The approach focuses on improving visual observability by enhancing contrast, emphasizing structural details, and controlling noise, while maintaining suitability for deployment in resource-constrained systems.

Based on the analysis of related methods, a processing pipeline was developed using multi-scale decomposition combined with selective detail enhancement. By separating the base layer and detail components, the method enables flexible control over the enhancement process, resulting in improved image quality and stable visual representation. Experimental results demonstrate that the method effectively enhances thermal images, particularly in low-contrast and noisy conditions. At the same time, certain limitations remain, including sensitivity to parameter selection and the computational cost of specific processing stages.

Future work may focus on improving adaptability through the integration of learning-based components, as well as optimizing computational performance to support real-time processing. In addition, further evaluation on more diverse datasets is necessary to validate the robustness and applicability of the method across different scenarios.

Overall, the study achieves its intended objectives and presents a practical and effective approach for thermal image enhancement in real-world observation systems.

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APPENDIX

Appendix A: Additional Experimental Results

Figure A.1: Input images and corresponding enhanced thermal results

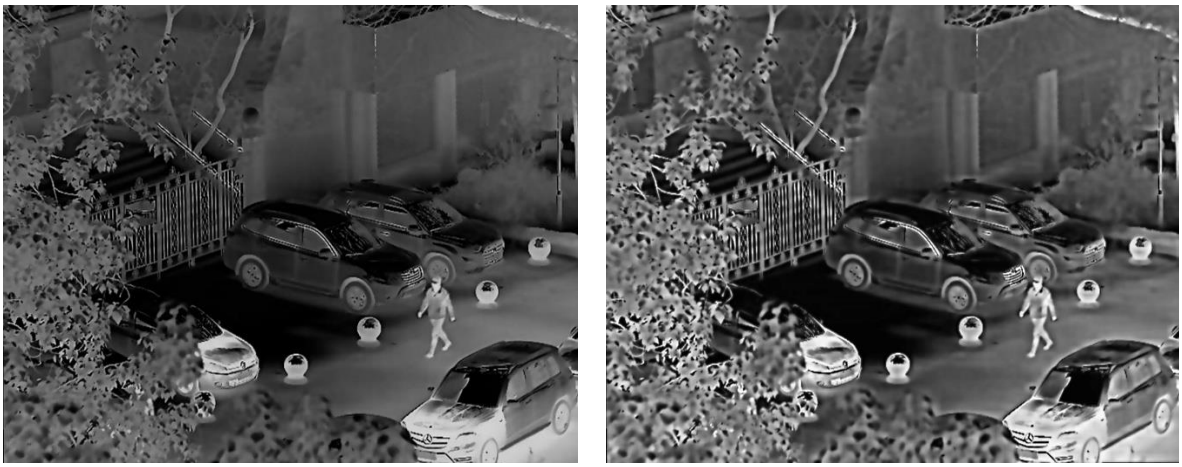

Figure A.2: Results of thermal–visible image fusion

Figure A.3: Visual comparison between different enhancement methods

Image	Entropy_RGF	Entropy_Hybrid	Entropy_MSGF	Laplacian_RGF	Laplacian_Hybrid	Laplacian_MSGF	Sobel_RGF	Sobel_Hybrid	Sobel_MSGF	RMS_RGF	RMS_Hybrid	RMS_MSGF
90811	7.087645	7.220749	7.367502	6.613179	31.394309	22.226872	1080.515192	2479.065564	1889.158768	43.829571	47.159721	47.189722
90812	7.377193	7.419484	7.478223	3.315256	22.673246	15.135887	529.308664	1664.861438	1251.032008	44.636946	45.284411	46.813539
90813	7.207771	7.279654	7.373778	3.143709	21.2629	14.950887	429.165361	1382.503934	1011.354356	42.149218	43.685662	44.287697
90814	7.15264	7.24244	7.364072	5.233624	28.027081	19.841254	790.055182	2044.117014	1505.442014	44.533001	46.451484	46.900335
90815	7.343641	7.386828	7.507086	2.773457	18.880153	12.13257	432.615204	1312.920177	941.77782	44.214178	45.038669	47.418536
90816	7.276521	7.381008	7.473816	5.63764	29.149601	19.886192	883.145068	2224.68313	1586.57744	45.325134	47.855059	49.011124
90817	7.115267	7.247615	7.414101	6.190796	29.737355	20.8548	1117.73432	2460.831151	1874.509416	44.296101	47.598435	47.963234
90819	7.135396	7.275924	7.419111	4.495176	24.662326	16.485001	809.824692	2060.125027	1447.894234	42.432331	45.832669	47.194369
90820	7.227051	7.347719	7.481932	5.195897	28.131275	18.729386	826.620026	2148.475578	1560.787213	45.636881	48.264996	49.537919
90821	7.295281	7.386967	7.493398	4.171304	25.112471	17.263767	610.994019	1815.466353	1371.587898	45.37725	47.504514	48.944762
90822	7.257579	7.361932	7.487743	4.756983	26.422181	17.697718	707.529883	1947.863861	1453.027254	45.709843	47.87929	49.318364
90823	7.326177	7.416299	7.526691	4.197773	24.954422	16.932465	643.074225	1887.952745	1438.086342	46.801357	48.660929	50.123696
90824	7.350878	7.409564	7.497342	3.574401	21.885001	14.284145	493.013882	1454.893835	1036.339569	46.625819	48.134234	49.19331
90825	7.213957	7.26195	7.374098	2.374491	17.686485	11.297694	328.458676	1153.059973	787.218292	42.679598	43.814345	44.530772
90826	7.237821	7.349194	7.488286	5.454411	28.86099	19.678002	900.142059	2297.729025	1629.715799	45.032986	47.881165	49.667003
90827	7.169203	7.299016	7.428255	7.579477	35.980091	25.638271	1442.330859	3239.514713	2449.061507	43.98755	47.642549	48.208747
90828	7.198642	7.310531	7.471267	3.82626	22.26473	15.269704	640.291937	1736.686122	1283.286565	44.446583	47.323117	48.819443
90829	7.380083	7.41931	7.511042	2.885793	19.90834	13.089146	501.566156	1581.411961	1163.024033	44.823313	45.72956	47.659173
90830	6.920201	7.095947	7.294222	7.13578	32.620298	24.313692	1402.978107	2984.920734	2203.994072	39.787761	44.75387	45.463835
90831	7.222793	7.317451	7.452251	5.453219	28.961832	20.572253	872.344258	2267.57218	1693.891588	44.526162	46.926303	48.821041
90832	7.110796	7.210086	7.342704	6.002029	29.386401	20.708655	1011.277284	2282.847252	1649.989157	40.738434	43.078515	43.719691
90833	7.27323	7.328104	7.428958	3.106119	21.00729	14.276107	455.410393	1442.766191	1038.702057	43.996305	45.044124	45.956373
90834	7.312652	7.350138	7.457866	2.940572	19.819558	13.996813	472.988745	1366.628389	1015.787779	42.978344	43.949858	45.875782
90835	7.195889	7.311861	7.444436	5.037928	27.183399	18.578381	790.264076	2065.715805	1571.078955	44.84117	47.511271	48.344884
90836	6.506396	6.680056	6.902745	4.670163	21.359828	13.372395	1029.111386	2058.116698	1434.211957	38.920953	44.512925	41.717493
90837	7.210139	7.317582	7.473346	7.540089	36.131824	24.238863	1331.873825	2990.129256	2229.502679	43.699406	46.186004	48.14149
90838	6.53926	6.738135	6.969315	15.215408	57.856483	35.435469	2729.194861	5193.397127	3570.091888	40.766545	47.831579	44.22407
90839	7.288381	7.383473	7.522711	3.71659	22.515175	14.86513	586.624133	1750.466112	1224.00712	45.904036	47.915899	49.879991
90840	7.314994	7.404676	7.487894	3.514452	21.707435	14.519136	490.24299	1537.509978	1109.366495	46.565717	48.174955	49.376564
90841	7.093466	7.199209	7.341731	3.392993	20.040844	13.442803	523.770206	1386.590179	993.088118	40.03238	42.985341	44.978205
90842	7.296278	7.403746	7.525661	3.836082	23.598009	15.855126	575.641158	1756.655423	1315.405042	46.3569	48.386782	49.986374
90843	6.592105	6.803221	7.039885	14.57129	53.609807	35.067441	2673.345947	4962.053532	3511.097263	41.064851	47.807176	45.079785
90844	7.216483	7.29296	7.423847	3.314394	19.805298	13.157022	519.996219	1365.111995	983.686465	43.512867	45.880836	47.389998

Appendix B: System Interface

Figure B.1: Main interface of the implemented system

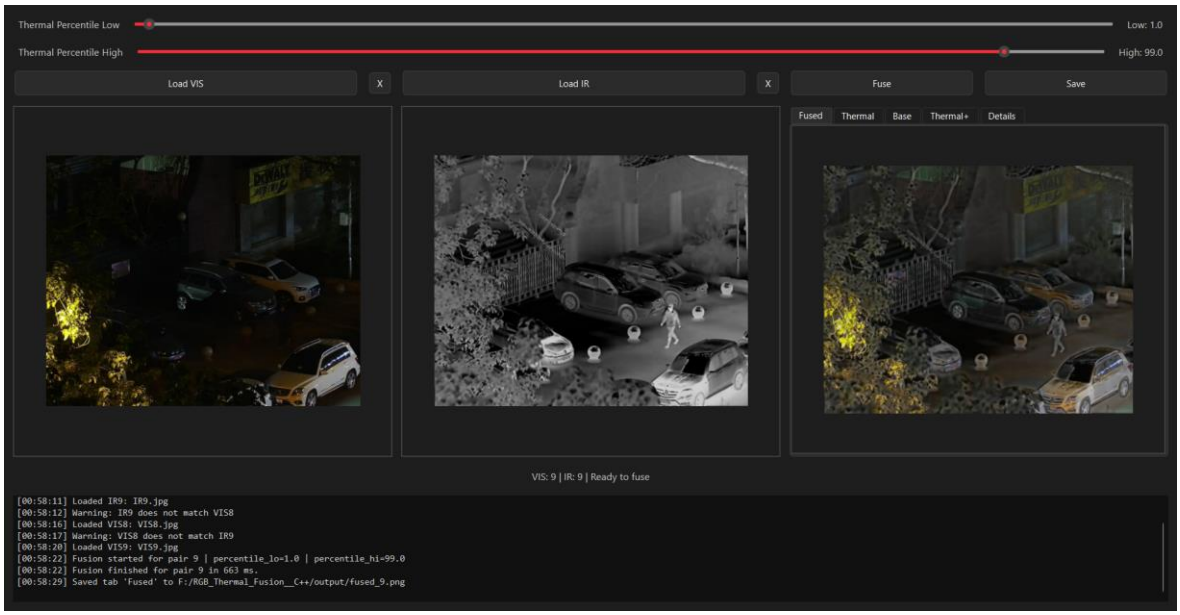
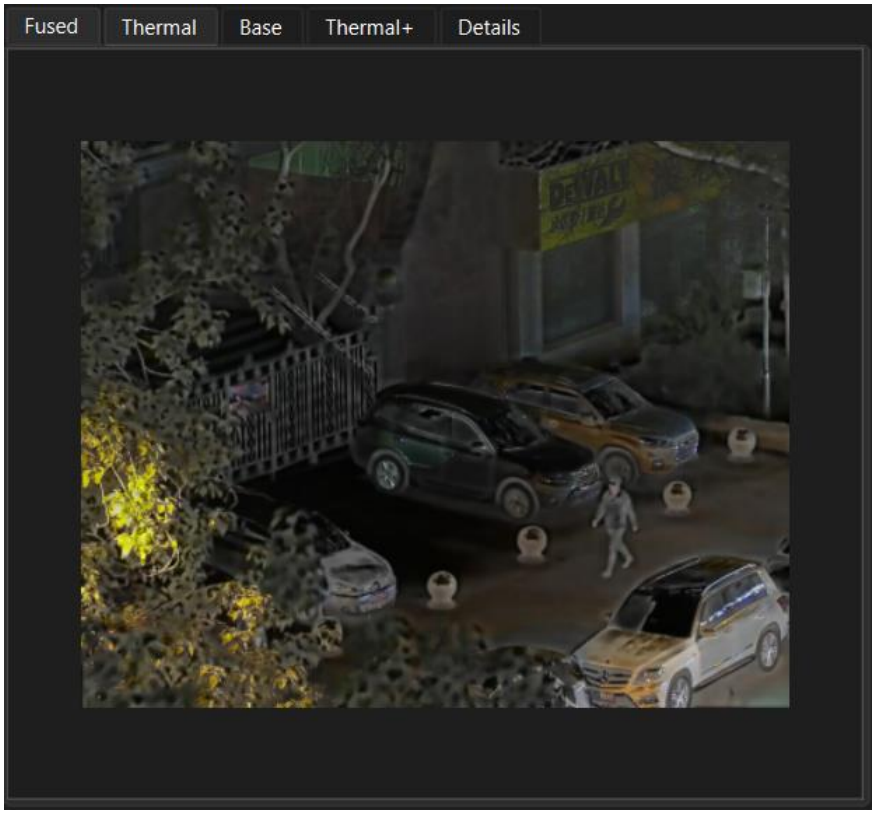
Figure B.2: Tab-based visualization of intermediate and final results